Blaschke products associated to ellipses: a simple characterization

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Abstract

We study finite Blaschke products whose level sets on the unit circle are the vertices of polygons circumscribing a fixed ellipse (Poncelet polygons). Theorem X describes such a Blaschke product through the foci of a family of ellipses. Theorem Y gives the elliptic curve hidden behind the configuration. Theorem Z, the main new result, characterizes these Blaschke products through their critical values only: B is associated to an ellipse if and only if it has one critical value in the disk, or exactly two with simple critical points. The proof is elementary: follow the vertices while the parameter travels around the critical values, and check whether neighbors stay neighbors. Equivalently, B is associated to an ellipse if and only if its monodromy group is a group of symmetries of the polygon, and this group is then cyclic or dihedral. Behind it lies an isogeny between two elliptic curves, of which such a B is the "shadow". Section 2 is a self-contained introduction to the objects involved (Riemann surfaces, elliptic functions, elliptic curves, isogenies), written for readers who have never met them.

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1 Introduction

Let $\mathbb{D}$ be the open unit disk and $\Gamma = \partial\mathbb{D}$ the unit circle. A *finite Blaschke product* of degree n is a rational function

$$B(z) = e^{i\theta} \prod_{j=1}^{n} \frac{z - a_j}{1 - \bar{a}_j z}, \qquad a_j \in \mathbb{D}.$$

It maps $\mathbb{D}$ onto $\mathbb{D}$ and Γ onto Γ, and every $\lambda \in \Gamma$ has exactly n distinct preimages on Γ. Following Daepp, Gorkin, Shaffer and Voss [3], we say that B is *normalized* if $B(0) = 0$ and $e^{i\theta} = 1$, i.e. $B(z) = z \prod_{j=1}^{n-1} \frac{z-a_j}{1-\bar{a}_j z}$. We lose no generality by studying normalized Blaschke products (see Remark 7).

Definition 1. A Blaschke product B of degree $n \geq 3$ is *associated* to a curve $\gamma \subset \mathbb{D}$ if, for every $\lambda \in \Gamma$, the n solutions of $B(z) = \lambda$, taken in counterclockwise order, are the vertices of a polygon whose sides are all tangent to γ.

For every normalized B, the polygons of Definition 1 circumscribe a convex curve, the *Blaschke curve* of B, which is the boundary of the numerical range of a compressed shift operator; see [3] and [1, §3]. The question is when this curve is an ellipse. For degree 3, it always is, and its foci are the two non-zero zeros of B (Daepp, Gorkin and Mortini [2]). For degree 4, it is an ellipse if and only if B is a composition of two Blaschke products of degree 2 (Fujimura [4]; see Gorkin and Wagner [5] for an operator-theoretic proof, and [3, Cor. 13.9]). Degree 6 is treated by Hunziker, Martínez-Finkelshtein, Poe and Simanek [6]. Aksoy, Arici, Celorrio and Gorkin [1] showed that an elliptical Blaschke curve forces B to be decomposable along every factorization of its degree, that a Blaschke product with a single critical value always has an elliptical Blaschke curve, and that in degree 2^n an elliptical Blaschke curve allows at most n distinct critical values. Critical values also appear in the decomposition algorithms of [7].

In this article we give a necessary and sufficient condition in every degree, in terms of critical values only (Theorem Z): *B is associated to an ellipse if and only if it has one critical value in $\mathbb{D}$, or exactly two with simple critical points.* The proof follows the vertices of a polygon while λ goes around the critical values of B: the ellipse exists exactly when neighboring vertices always come back as neighbors, i.e. when the monodromy group of B is contained in the dihedral group of the polygon; it is then cyclic in case (a) and dihedral in case (b) (Theorem Z′ in §5.2). The underlying reason is that such a B is the shadow, on the Riemann sphere, of an isogeny between two elliptic curves. That elliptic curves govern Poncelet's porism is classical [12, 11]; see also [10] for Poncelet polygons inscribed in a circle.

Organization. Section 2 is a crash course on the tools used later. Section 3 contains Theorem X (zeros of B = foci of a family of ellipses), Section 4 contains Theorem Y (the elliptic curve of the configuration), Section 5 contains Theorem Z (characterization through critical values), the formula of Corollary 1, and a comparison with the results of [1]. Section 6 presents explicit examples of degrees 5, 4 and 8, with their ellipses.

Reading advice. A reader who knows what an isogeny of elliptic curves is can skip Section 2. The others should read it: every statement used later is there, with an explanation of why it is true. The proof of Theorem Z (§5.1 and §5.2) uses only §2.1, §2.6, §2.7, Lemma 1 and Lemma 6; elliptic curves come back in §5.4.

2 A crash course for non-specialists

This section collects the facts we need, with an explanation of why each is true. For complete treatments, see [13] for Riemann surfaces and [15] for elliptic curves and isogenies.

2.1 Rational maps, local degree, critical points

The *Riemann sphere* is $\mathbb{P}^1 = \mathbb{C} \cup \{\infty\}$. A rational function $R = N/D$ (with N, D coprime polynomials) defines a map $R\colon \mathbb{P}^1 \to \mathbb{P}^1$ whose *degree* is $d = \max(\deg N, \deg D)$: every $w \in \mathbb{P}^1$ has exactly d preimages, *counted with multiplicity*.

The multiplicity is the *local degree*. Near a point z_0 (with $R(z_0)$ finite) we can write $R(z) - R(z_0) = c\,(z - z_0)^e + \dots$ with $c \neq 0$; the integer $e = e_R(z_0) \geq 1$ is the local degree of R at z_0. Near z_0, the map R behaves like $u \mapsto u^e$: it is e-to-1 around z_0.

- If $e_R(z_0) \geq 2$, then z_0 is a *critical point* and $R(z_0)$ a *critical value*. The critical point is *simple* if $e_R(z_0) = 2$. Equivalently, $R'(z_0) = 0$, and z_0 is simple iff moreover $R''(z_0) \neq 0$.
- Counting the preimages of w without multiplicity:

$$\#R^{-1}(w) = d - \sum_{z \in R^{-1}(w)} \big(e_R(z) - 1\big). \tag{1}$$

Example 1. $R(z) = z^n$ has degree n. Its only critical points are 0 and ∞, both with local degree n. The fibre of $w \neq 0, \infty$ is the set of n distinct n-th roots of w; the fibre of 0 is the single point 0, in agreement with (1): $n - (n-1) = 1$.

Fact 2.1 (Blaschke products)**.** *Let B be a Blaschke product of degree n.*

1. *$B(1/\bar{z}) = 1/\overline{B(z)}$. Hence the critical points (and values) of B are symmetric with respect to Γ under $z \mapsto 1/\bar{z}$.*
2. *B has no critical point on Γ, exactly $n-1$ critical points in $\mathbb{D}$ (counted with multiplicity) and $n-1$ outside $\overline{\mathbb{D}}$ (Walsh's theorem; see [1, Thm. 4.1]).*
3. *B maps $\mathbb{D}$ to $\mathbb{D}$ and $\mathbb{P}^1 \setminus \overline{\mathbb{D}}$ to itself, so the fibre of $v \in \mathbb{D}$ lies in $\mathbb{D}$, and (1) can be applied inside $\mathbb{D}$.*
4. *$B|_\Gamma\colon \Gamma \to \Gamma$ is an n-to-1 covering which preserves the orientation: as z goes once counterclockwise around Γ, $B(z)$ goes n times counterclockwise around Γ.*

2.2 Riemann surfaces, genus, and the Riemann–Hurwitz formula

A *compact Riemann surface* is a compact surface on which one can do complex analysis: locally it looks like an open set of $\mathbb{C}$, and changes of coordinates are holomorphic. Examples: the sphere $\mathbb{P}^1$; the tori $\mathbb{C}/\Lambda$ of §2.3; smooth algebraic curves. Topologically, a compact Riemann surface is a sphere with g handles; g is the *genus*. The sphere has genus 0, a torus has genus 1.

All the notions of the previous subsection (degree of a map, local degree, critical points, formula (1)) make sense for a non-constant holomorphic map $h\colon S \to S'$ between compact Riemann surfaces. They are related to the genera by:

Fact 2.2 (Riemann–Hurwitz formula)**.** *If $h\colon S \to S'$ has degree d, then*

$$2g(S) - 2 = d\big(2g(S') - 2\big) + \sum_{p \in S} \big(e_h(p) - 1\big).$$

We will use it three times.

- For a rational map $R\colon \mathbb{P}^1 \to \mathbb{P}^1$ of degree d: $-2 = -2d + \sum(e-1)$, so R has $2d-2$ critical points counted with multiplicity.
- *A double cover of the sphere branched over 4 points has genus 1.* If $h\colon S \to \mathbb{P}^1$ has degree 2 and exactly 4 points with $e_h = 2$, then $2g(S) - 2 = 2(-2) + 4 = 0$, so $g(S) = 1$. The typical example is the curve $y^2 = \Delta(x)$, where Δ is a polynomial of degree 3 or 4 with distinct roots, and $h(x, y) = x$: above each x there are two points $(x, \pm y)$, except above the roots of Δ (and above ∞ if $\deg \Delta = 3$).
- *A map between two surfaces of genus 1 has no critical point.* Indeed $0 = d \cdot 0 + \sum(e_h(p) - 1)$ forces $e_h(p) = 1$ everywhere. Such a map is said to be *unramified*.

2.3 Complex tori and elliptic functions

Let $\omega_1, \omega_2 \in \mathbb{C}$ be linearly independent over $\mathbb{R}$, and $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ the *lattice* they generate. The quotient $E = \mathbb{C}/\Lambda$ is obtained from the parallelogram with vertices $0, \omega_1, \omega_2, \omega_1 + \omega_2$ by gluing opposite sides: it is a torus (Figure 1). It is a compact Riemann surface of genus 1, and it is also a *group*: we add points of E by adding complex numbers modulo Λ.

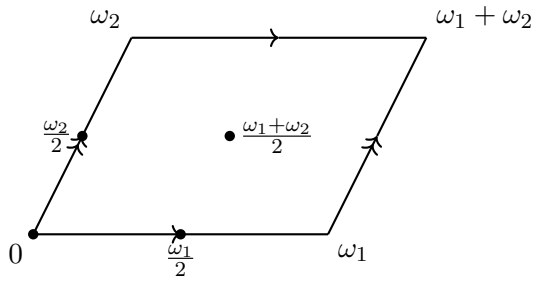


Figure 1: The torus $E = \mathbb{C}/\Lambda$: sides with the same arrows are glued. The four marked points form $E[2]$, the points t with $2t \in \Lambda$.

Definition 2. An *elliptic function* for Λ is a meromorphic function f on $\mathbb{C}$ such that $f(t+\omega) = f(t)$ for all $\omega \in \Lambda$. Equivalently, it is a meromorphic function on the torus E, i.e. a holomorphic map $f\colon E \to \mathbb{P}^1$.

The name comes from the computation of the arc length of an ellipse, which leads to the integrals $\int dx/\sqrt{\Delta(x)}$ of Section 4. The analogy to keep in mind is $\int dx/\sqrt{1-x^2} = \arcsin x$: the inverse function, sin, is periodic with *one* period. When $\sqrt{1-x^2}$ is replaced by the square root of a polynomial of degree 3 or 4, the inverse function has *two* independent periods: it is an elliptic function.

Fact 2.3. *Let f be a non-constant elliptic function.*

1. *f has poles: an elliptic function without poles is bounded on the parallelogram, hence on $\mathbb{C}$, hence constant (Liouville's theorem).*
2. *As a map $E \to \mathbb{P}^1$, f has a degree d (every value is taken d times, with multiplicity), and $d \geq 2$.*
3. *The classical example is the Weierstrass function $\wp$, which has degree 2 and is even: $\wp(-t) = \wp(t)$.*

Even functions of degree 2 ("X-coordinates"). Suppose f is elliptic, even, of degree 2. For each t, the two points t and $-t$ have the same image, and since the degree is 2 there is nothing else: the fibres of f are the pairs $\{t, -t\}$. These two points coincide exactly when $2t \in \Lambda$, that is, at the four points

$$E[2] = \left\{0,\ \tfrac{\omega_1}{2},\ \tfrac{\omega_2}{2},\ \tfrac{\omega_1+\omega_2}{2}\right\}$$

of Figure 1. So *f has exactly 4 critical points, the points of $E[2]$, each of local degree 2*, and 4 critical values. In other words, f realizes E as a double cover of the sphere branched over 4 points, exactly as in the second use of Riemann–Hurwitz. Conversely:

Fact 2.4 (Uniformization)**.** *Every compact Riemann surface of genus 1 is isomorphic to a torus $\mathbb{C}/\Lambda$. For the curve $y^2 = \Delta(x)$ the isomorphism is explicit: the map $(x, y) \mapsto t = \int_{x_0}^{x} \frac{dx}{y}$ (the integral being taken along a path on the curve) is well defined modulo a lattice Λ of periods, and its inverse is $x = f(t)$, $y = f'(t)$, where f is an elliptic function of degree 2, even after a translation of t.*

This is why the coordinate x is called the X-coordinate, and y the Y-coordinate, of the elliptic curve; Theorems X and Y owe their names to this.

2.4 Translations and involutions of a torus

For $c \in \mathbb{C}$ the *translation* $t \mapsto t + c$ is an automorphism of E. It has no fixed point unless $c \in \Lambda$ (in which case it is the identity). The map $t \mapsto c - t$ is also an automorphism: a "half-turn". Its fixed points are the solutions of $2t \equiv c$; there are exactly 4 of them.

Fact 2.5. *Let $j\colon E \to E$ be a holomorphic involution ($j \circ j = \mathrm{id}$, $j \neq \mathrm{id}$) having at least one fixed point. Then $j(t) = c - t$ for some c. If moreover $j(0) = 0$, then $j(t) = -t$.*

Why. Every automorphism of E lifts to $\mathbb{C}$ as $t \mapsto ut + c$ with $u\Lambda = \Lambda$ (its derivative is an elliptic function without poles, hence constant). From $j \circ j = \mathrm{id}$ we get $u^2 = 1$. If $u = 1$, j is a translation, without fixed points. So $u = -1$. □

Fact 2.6 (Two half-turns make a translation)**.** *If $j_1(t) = c_1 - t$ and $j_2(t) = c_2 - t$, then $j_2 \circ j_1(t) = t + (c_2 - c_1)$ is a translation.*

This elementary fact is the heart of Poncelet's theorem (Lemma 5).

Torsion points. A point $\omega \in E$ has *order* n if $n\omega \in \Lambda$ and $k\omega \notin \Lambda$ for $0 < k < n$. Then the translation by ω, iterated n times, is the identity. The points of order dividing m form a group $E[m] \cong (\mathbb{Z}/m\mathbb{Z})^2$ with m^2 elements.

2.5 Isogenies

Definition 3. Let $\Lambda \subset \Lambda'$ be two lattices. The map

$$\pi\colon E = \mathbb{C}/\Lambda \longrightarrow E' = \mathbb{C}/\Lambda', \qquad t \bmod \Lambda \longmapsto t \bmod \Lambda'$$

is called an *isogeny*. Its *kernel* is the finite group $K = \Lambda'/\Lambda \subset E$.

The isogeny π "forgets" the difference between points that differ by an element of K:

- π is a group homomorphism: $\pi(t+s) = \pi(t) + \pi(s)$ and $\pi(-t) = -\pi(t)$;
- the fibres of π are the cosets $t + K$, so $\deg \pi = |K|$;
- π is unramified: in the coordinate t it is the identity, so $e_\pi = 1$ everywhere.

Example 2 (The isogeny of a Poncelet configuration)**.** Let $\omega \in E$ be a point of order n and $\Lambda' = \Lambda + \mathbb{Z}\omega$. Then K is the cyclic group $\{0, \omega, 2\omega, \ldots, (n-1)\omega\}$, and $\pi\colon E \to E'$ identifies all the points of an orbit t, $t+\omega$, $\ldots$, $t+(n-1)\omega$ of the translation by ω. In Section 5 the points of such an orbit will be the vertices of a Poncelet polygon, and π will be the elliptic counterpart of B, which also identifies the vertices of a polygon.

Fact 2.7. *Let $h\colon E \to E'$ be a non-constant holomorphic map between two tori with $h(0) = 0$. Then h is unramified (Riemann–Hurwitz), it lifts to $\mathbb{C}$ as $t \mapsto$ at with $a\Lambda \subset \Lambda'$, it is a group homomorphism, and its fibres are the cosets of its kernel K, a finite group with $|K| = \deg h$. Up to the harmless change of coordinate $t \mapsto$ at, h is an isogeny in the sense of the definition above.*

The kernel of an isogeny need not be cyclic: for the multiplication $t \mapsto mt$ it is $E[m] \cong (\mathbb{Z}/m)^2$. The isogeny attached to a Poncelet configuration has a cyclic kernel, generated by the "next vertex" translation (§5.4).

2.6 Monodromy

Let $h\colon S \to S'$ be a holomorphic map of degree d and ℓ a closed loop in S', based at w, avoiding the critical values of h. Each of the d points p of $h^{-1}(w)$ is the starting point of a unique path in S lying above ℓ (a *lift* of ℓ); its endpoint is again in $h^{-1}(w)$. This defines a permutation of the fibre $h^{-1}(w)$: the *monodromy* of ℓ.

Example 3. For $z \mapsto z^n$ and the loop Γ, the lift starting at a root z of $z^n = w$ ends at $e^{2i\pi/n}z$: the monodromy is a cyclic permutation of the n roots (an *n-cycle*). More generally, by Fact 2.1(4), for a Blaschke product B and the loop Γ, the monodromy sends each solution of $B(z) = \lambda$ to the next one counterclockwise: it is an n-cycle.

2.7 Duality of conics

A line $L_x x + L_y y + L_c = 0$ is represented by the vector (L_x, L_y, L_c), up to a non-zero scalar. If a non-degenerate conic has equation $x^T A x = 0$ (with $x = (x, y, 1)$), its tangent lines are exactly the lines L with $L^T A^{-1} L = 0$: the tangent lines form the *dual conic*, of matrix A^{-1}. Conversely, a matrix M such that the lines L with $L^T M L = 0$ are the tangents of a curve determines the conic $A = M^{-1}$. We will use *Bézout's theorem* in its simplest form: two distinct irreducible conics have at most 4 common points (also when the "points" are lines of a dual conic).

3 Theorem X: zeros and foci

3.1 Hypotheses and notation

We study the Poncelet porism configuration where the outer conic Γ is the unit circle and the inner conic γ is an ellipse strictly contained in $\mathbb{D}$. Let $x^T Ax = 0$ be the matrix equation of γ, where

$$A = \begin{pmatrix} a & b/2 & d/2 \\ b/2 & c & e/2 \\ d/2 & e/2 & f \end{pmatrix}, \qquad x = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix},$$

with $a, b, c, d, e, f \in \mathbb{R}$. The equation of the dual γ' of γ is $L^T A^{-1} L = 0$ (§2.7).

Let us connect this equation with Γ. Let L be a line tangent to γ, and $\varphi, \psi \in \Gamma$ its two intersections with Γ. Writing $z = x + iy$, the line through φ and ψ is $z + \varphi\psi\bar{z} = \varphi + \psi$, i.e.

$$\mathbf{d}(\varphi, \psi) = \big(i(\varphi\psi + 1),\ \varphi\psi - 1,\ -i(\varphi + \psi)\big).$$

(We write $\mathbf{d}$ in bold to distinguish it from the coefficient d of A.) Substituting into the equation of γ' gives

$$\mathbf{d}(\varphi, \psi)^T A^{-1} \mathbf{d}(\varphi, \psi) = 0. \tag{2}$$

Remark 1. Equation (2) can be seen as an algebraic relation between two consecutive vertices φ and ψ of a polygon circumscribing γ, instead of a relation between the coefficients of the tangent lines.

Remark 2. For $|\varphi| = |\psi| = 1$, one checks $\overline{\mathbf{d}(\varphi, \psi)} = -\bar{\varphi}\bar{\psi}\,\mathbf{d}(\varphi, \psi)$: the vector $\mathbf{d}(\varphi, \psi)$ is real up to a scalar, as it should be for a real line. For arbitrary complex φ, ψ one has $\overline{\mathbf{d}(\varphi, \psi)} = -\bar{\varphi}\bar{\psi}\,\mathbf{d}(1/\bar{\varphi}, 1/\bar{\psi})$.

3.2 The biquadratic relation

Lemma 1. *Equation* (2) *is equivalent to* $F(\varphi, \psi) = 0$, *where*

$$\begin{aligned} F(\varphi, \psi) = {} & \psi^2\big[\varphi^2(a_6 - a_3 - ia_4) + \varphi(a_5 + ia_2) - a_1\big] \\ & + \psi\big[\varphi^2(a_5 + ia_2) - 2(a_1 + a_3 + a_6)\varphi + a_5 - ia_2\big] \\ & - a_1\varphi^2 + (a_5 - ia_2)\varphi + a_6 - a_3 + ia_4, \end{aligned}$$

with

$$a_1 = b^2 - 4ac, \quad a_2 = 2bd - 4ae, \quad a_3 = e^2 - 4cf, \quad a_4 = 2de - 4bf, \quad a_5 = -2be + 4cd, \quad a_6 = d^2 - 4af.$$

Proof. One checks (by computing the adjugate of A) that

$$A^{-1} = \frac{-1}{4\det(A)} \begin{pmatrix} a_3 & -a_4/2 & a_5/2 \\ -a_4/2 & a_6 & -a_2/2 \\ a_5/2 & -a_2/2 & a_1 \end{pmatrix}.$$

We can ignore the factor $-1/(4 \det A)$. For $\mathbf{d} = (d_x, d_y, d_c)$ we get

$$E(d_x, d_y, d_c) = a_1 d_c^2 + a_3 d_x^2 + a_6 d_y^2 - a_2 d_c d_y - a_4 d_y d_x + a_5 d_c d_x.$$

Replacing d_x, d_y, d_c by their expressions in φ, ψ:

$$\begin{aligned} E(\varphi, \psi) = {} & -a_1(\varphi + \psi)^2 - a_3(\varphi\psi + 1)^2 + a_6(\varphi\psi - 1)^2 \\ & + ia_2(\varphi + \psi)(\varphi\psi - 1) - ia_4(\varphi\psi - 1)(\varphi\psi + 1) + a_5(\varphi\psi + 1)(\varphi + \psi). \end{aligned}$$

Collecting the coefficients of ψ^2, ψ and 1 gives exactly $F(\varphi, \psi)$. □

From now on we write

$$F(\varphi,\psi)=P(\varphi)\psi^2+Q(\varphi)\psi+R(\varphi),\qquad \Delta(\varphi)=Q(\varphi)^2-4P(\varphi)R(\varphi).$$

Remark 3. F is symmetric in φ and ψ, because (2) obviously is. If we fix $\varphi=u\in\Gamma$, the two solutions ψ of $F(u,\psi)=0$ are the second intersections with Γ of the two tangents to γ through u. They coincide exactly when $\Delta(u)=0$, i.e. when the two tangents through u coincide, i.e. when $u\in\gamma$. (For $u\in\Gamma$ the leading coefficient $P(u)$ does not vanish, by Lemma 2 below.) More generally, a point $u\in\mathbb{P}^1$ parametrizes the point $\left(\frac{u+u^{-1}}{2},\frac{u-u^{-1}}{2i}\right)$ of the complexified circle, and $\Delta(u)=0$ exactly when this point lies on the complexified conic γ. Since $\gamma\cap\Gamma=\emptyset$, no root of Δ lies on Γ: the roots of Δ (four in $\mathbb{P}^1$, counted with multiplicity) are the four complex, non-real, intersection points of Γ and γ.

3.3 Foci

Lemma 2. *Let α and β be the foci of γ, seen as complex numbers. Then*

$$R(\varphi)=-a_1(\varphi-\alpha)(\varphi-\beta),\qquad P(\varphi)=-a_1(1-\bar\alpha\varphi)(1-\bar\beta\varphi),$$

and $F=0$ can be rewritten as

$$(1-\bar\alpha\varphi)(1-\bar\beta\varphi)\psi^2-\left[(\bar\alpha+\bar\beta)\varphi^2+\sigma\varphi+(\alpha+\beta)\right]\psi+(\varphi-\alpha)(\varphi-\beta)=0,\qquad \sigma=-\frac{2(a_1+a_3+a_6)}{a_1}.\quad(3)$$

Proof. The key fact (**) is that the foci are the roots of $R(\varphi)=F(\varphi,0)$. Note that $\mathbf{d}(\varphi,0)=(i,-1,-i\varphi)$ is the line $ix-y-i\varphi=0$, i.e. the line through the point φ with direction $(1,i)$: an *isotropic line*. By a classical characterization due to Plücker, *the foci of a conic are the points through which the isotropic lines are tangent to the conic.* Let us check it directly. Both sides are compatible with translations and rotations (a rotation by θ multiplies φ by $e^{i\theta}$ and preserves isotropic directions), so we may assume γ is $x^2/p^2+y^2/q^2=1$, whose dual equation is $p^2L_x^2+q^2L_y^2=L_c^2$. For $L=(i,-1,-i\varphi)$ this reads $-p^2+q^2=-\varphi^2$, i.e. $\varphi=\pm\sqrt{p^2-q^2}$: these are the foci. Since $R(\varphi)=\mathbf{d}(\varphi,0)^TA^{-1}\mathbf{d}(\varphi,0)$ up to a constant, (**) follows.

The leading coefficient of R is $-a_1$, so $R(\varphi)=-a_1(\varphi-\alpha)(\varphi-\beta)$. The coefficients of P are those of R, conjugated and in reverse order, so $P(\varphi)=\varphi^2\overline{R(1/\bar\varphi)}=-a_1(1-\bar\alpha\varphi)(1-\bar\beta\varphi)$ (a_1 is real). The sum of the foci is $\alpha+\beta=(a_5-ia_2)/a_1$, which gives the middle coefficient. Divide by $-a_1$, which is non-zero ($a_1<0$ for an ellipse). □

Lemma 3. *Let $\varphi=z_0\in\Gamma$. Then the product of the two solutions $\psi=z_1$ and $\psi=z_{-1}$ of $F(z_0,\psi)=0$ is*

$$z_1z_{-1}=\frac{z_0-\alpha}{1-\bar\alpha z_0}\cdot\frac{z_0-\beta}{1-\bar\beta z_0}.$$

Proof. The product of the roots of ax^2+bx+c is c/a; apply it to (3). □

We already see a fragment of a Blaschke product.

3.4 Diagonals

Lemma 4. *Let P be an n-gon inscribed in Γ, with consecutive vertices $z_0,\dots,z_{n-1}$, which circumscribes γ (so that, by Poncelet's porism, there is such a polygon through every point of Γ). For $k\in\{1,\dots,n-1\}$, let P_k be the polygon with sides z_iz_{i+k} (indices mod n).*

1. *If $2k \not\equiv 0 \pmod n$, the polygons P_k all circumscribe one and the same ellipse $\gamma_k \subset \mathbb{D}$. Note that $\gamma_k = \gamma_{n-k}$ and $\gamma_1 = \gamma$.*

2. *If n is even and $k = n/2$, all the long diagonals $z_i z_{i+n/2}$ (of all the polygons) pass through one point $P_{n/2} \in \mathbb{D}$, and $z_{i+n/2} = \dfrac{P_{n/2} - z_i}{1 - \bar{P}_{n/2} z_i}$.*

This is a classical theorem of Darboux; see [1, Thm. 2.1] and [9], where the family $\gamma_1, \gamma_2, \ldots$ is called the *Poncelet package*. A self-contained proof, using the elliptic curve of the configuration, is given in §4.1. We write A_k for the matrix of γ_k, F_k for its biquadratic polynomial, and α_k, β_k for its foci.

3.5 Theorem X

Theorem X (Zeros of B are foci)**.** *Let $\gamma \subset \mathbb{D}$ be an ellipse admitting circumscribed n-gons inscribed in Γ, and let $B(z) = z\prod_{j=1}^{n-1} \frac{z-a_j}{1-\bar{a}_j z}$ be a normalized Blaschke product of degree n. Write $m = \lfloor (n-1)/2 \rfloor$. The following are equivalent:*

1. *B is associated to γ;*

2. *the zeros of B are 0, the foci α_k, β_k of $\gamma_1, \ldots, \gamma_m$, and, if n is even, the point $P_{n/2}$ of Lemma 4(2) (with multiplicities);*

3. *with $q_k(z) = \mathbf{d}(z,0)^T A_k^{-1} \mathbf{d}(z,0)$ and $q_k^*(z) = z^2\,\overline{q_k(1/\bar{z})}$, and a suitable constant $|e^{i\theta}| = 1$,*

$$B(z) = e^{i\theta} z \prod_{k=1}^{m} \frac{q_k(z)}{q_k^*(z)} \quad \times \quad \frac{z - P_{n/2}}{1 - \bar{P}_{n/2} z} \quad \textit{(this last factor only if } n \textit{ is even).}$$

Elliptic function form. *When Δ has four distinct roots, with the elliptic function f and the step ω of Lemma 5 below, a normalized B associated to γ satisfies, for all t,*

$$B\big(f(t)\big) = (-1)^{n+1} \prod_{j=0}^{n-1} f(t + j\omega).$$

This solves explicitly the equation $B(z) = \lambda$ for every $\lambda \in \Gamma$.

Remark 4. For $n = 3$, item 2 is the theorem of Daepp, Gorkin and Mortini [2]. In general, the fact that the zeros of B are foci of the ellipses of the Poncelet package is visible in the example of [1, Ex. 3.6]; on foci of ellipses inscribed in cyclic polygons, see also [6].

Proof. Items 2 and 3 are the same statement. By the proof of Lemma 2 applied to γ_k, $q_k(z) = \kappa_k(z-\alpha_k)(z-\beta_k)$ and $q_k^*(z) = \bar{\kappa}_k(1-\bar{\alpha}_k z)(1-\bar{\beta}_k z)$, so q_k/q_k^* is a Blaschke factor with zeros α_k, β_k, up to the unimodular constant $\kappa_k/\bar{\kappa}_k$.

The product computation. Recall that if B is normalized, the solutions $z_0, \ldots, z_{n-1}$ of $B(z) = \lambda$ are the roots of the monic polynomial $z\prod(z-a_j) - \lambda\prod(1-\bar{a}_j z)$, whose constant term is $-\lambda$; hence

$$\lambda = (-1)^{n+1} \prod_{i=0}^{n-1} z_i. \tag{4}$$

Now let $z_0, \ldots, z_{n-1}$ be the consecutive vertices of a polygon circumscribing γ, and B_0 the normalized Blaschke product whose zeros are listed in item 2. Suppose first that n is odd. For $1 \le k \le m$, the

vertices z_k and z_{-k} are the second intersections with Γ of the two tangents to γ_k through z_0 (they are sides of P_k through z_0). By Lemma 3 applied to γ_k,

$$z_k z_{-k} = \frac{(z_0-\alpha_k)(z_0-\beta_k)}{(1-\bar\alpha_k z_0)(1-\bar\beta_k z_0)}.$$

Multiplying over k:

$$\prod_{i=0}^{n-1} z_i = z_0 \prod_{k=1}^{m} z_k z_{-k} = B_0(z_0).$$

The left-hand side does not depend on which vertex we call z_0: *B_0 is constant on each polygon.* If n is even, the same computation works, using Lemma 4(2) for the last factor: $z_0 z_{n/2} = -z_0 \frac{z_0 - P_{n/2}}{1-\bar P_{n/2} z_0}$, and we get $\prod z_i = -B_0(z_0)$.

Since B_0 has degree n, its fibre over a point of Γ has exactly n points; it contains the n vertices of a polygon, hence is equal to it. So B_0 is associated to γ: this is $2 \Rightarrow 1$.

For $1 \Rightarrow 2$: if B is associated to γ, then B and B_0 have the same fibres on Γ: for $z \in \Gamma$, both fibres through z are the Poncelet polygon through z (which is unique: its vertices are obtained from z by drawing tangents). Let S be the image of $z \mapsto (B_0(z), B(z))$, an irreducible curve in $\mathbb{P}^1 \times \mathbb{P}^1$. For every $\lambda \in \Gamma$, the n points of $B_0^{-1}(\lambda)$ have one and the same image, so S has a single point above each $\lambda \in \Gamma$. The projection $S \to \mathbb{P}^1$, $(u,w) \mapsto u$, has a constant number of points in its fibres outside finitely many points; this number is therefore 1, so S is the graph of a rational function μ, and $B = \mu \circ B_0$. Comparing degrees, $\deg \mu = 1$ (see also the uniqueness statement [7, Thm. 2.3]). Since μ maps $\mathbb{D} = B_0(\mathbb{D})$ into $\mathbb{D}$ and $\mu(0) = \mu(B_0(0)) = B(0) = 0$, μ is a rotation; as both are normalized, $\mu = \mathrm{id}$ and $B = B_0$.

Elliptic form. By Lemma 5, the vertices of the polygon through $f(t)$ are the $f(t + j\omega)$; plug them into (4) with $\lambda = B(f(t))$. Both sides are elliptic functions of t which agree when $f(t) \in \Gamma$, hence everywhere. □

4 Theorem Y: the elliptic curve of the configuration

Let $\mathcal{C} = \{F(\varphi,\psi) = 0\} \subset \mathbb{P}^1 \times \mathbb{P}^1$: its points are the pairs of endpoints of chords tangent to γ (complex ones included). It is a curve of *bidegree* $(2,2)$: for each φ there are two values of ψ, and for each ψ two values of φ.

Theorem Y. *Assume Δ has four distinct roots (in $\mathbb{P}^1$). Then $\mathcal{C}$ is a compact Riemann surface of genus 1, isomorphic to the elliptic curve*

$$Y^2 = \Delta(X), \qquad \Delta(X) = \mathrm{disc}_\psi F(X,\psi) = Q(X)^2 - 4P(X)R(X),$$

via $(\varphi,\psi) \mapsto (X,Y) = \big(\varphi,\ 2P(\varphi)\psi + Q(\varphi)\big)$.

Proof. Completing the square, $F = 0$ is equivalent to $\big(2P(\varphi)\psi + Q(\varphi)\big)^2 = \Delta(\varphi)$, and the inverse map is $\psi = (Y - Q(X))/(2P(X))$: the two curves are birational, hence their smooth models are isomorphic (this is what "$\mathcal{C}$ is a Riemann surface" means here). Since Δ has simple roots, it is not a square, so $Y^2 = \Delta(X)$, hence F, is irreducible. The projection $(\varphi,\psi) \mapsto \varphi$ has degree 2 and is branched exactly over the four roots of Δ (where the two values of ψ coincide), so the genus is 1 by Riemann–Hurwitz. □

A second derivation. The following argument explains where the elliptic integral comes from. Write $F = A(\varphi)\psi^2 + B(\varphi)\psi + C(\varphi)$ (here A, B, C are polynomials, not the matrix and the Blaschke product). By symmetry also $F = A(\psi)\varphi^2 + B(\psi)\varphi + C(\psi)$. Differentiating $F(\varphi, \psi) = 0$ along the curve,

$$\big(2A(\varphi)\psi + B(\varphi)\big)\, d\psi + \big(2A(\psi)\varphi + B(\psi)\big)\, d\varphi = 0,$$

and since $2A(\varphi)\psi + B(\varphi) = \pm\sqrt{\Delta(\varphi)}$ and $2A(\psi)\varphi + B(\psi) = \mp\sqrt{\Delta(\psi)}$ (for a consistent choice of the square roots along the curve), this becomes

$$\frac{d\psi}{\sqrt{\Delta(\psi)}} = \frac{d\varphi}{\sqrt{\Delta(\varphi)}}.$$

In words: *in the coordinate $t = \int d\varphi/\sqrt{\Delta(\varphi)}$, the passage from a vertex to the next one is a translation $t \mapsto t+\omega$.* The inverse of this elliptic integral is the elliptic function $\varphi = f(t)$ of Fact 2.4, and $(f')^2$ is proportional to $\Delta(f)$: setting $X = f$, $Y = f'$ gives back the curve $Y^2 = \Delta(X)$ (up to a constant). The next lemma proves the same thing with Fact 2.6 instead of integrals.

Lemma 5 (Structure of the configuration)**.** *Assume Δ has four distinct roots, and write $\mathcal{C} = \mathbb{C}/\Lambda$ (Theorem Y and Fact 2.4). There is an elliptic function f and a point $\omega \in \mathcal{C}$ such that:*

1. *f is even and of degree 2; its critical points are the four points of $E[2]$, and its critical values are the four roots of Δ, i.e. the four complex points of $\Gamma \cap \gamma$;*
2. *the point of $\mathcal{C}$ with parameter t is $(\varphi, \psi) = \big(f(t), f(t+\omega)\big)$; in particular the vertices of the polygon through $f(t)$ are $f(t), f(t+\omega), f(t+2\omega), \ldots$;*
3. *the polygons close after n steps if and only if $n\omega \in \Lambda$; they have exactly n distinct vertices if and only if ω has order n.*

Proof. Consider the two involutions of $\mathcal{C}$

$$j\colon (\varphi, \psi) \mapsto (\varphi, \psi') \ \text{ (other root of } F(\varphi, \cdot)), \qquad s\colon (\varphi, \psi) \mapsto (\psi, \varphi).$$

j has four fixed points (the roots of Δ), and s has four fixed points (the points $\varphi = \psi$, which correspond to the four common tangents of Γ and γ). Choose the origin of $\mathcal{C}$ at a fixed point of j; by Fact 2.5, $j(t) = -t$ and $s(t) = c - t$. Think of a point (z_0, z_1) of $\mathcal{C}$ as the oriented side from z_0 to z_1. Then s sends it to (z_1, z_0) and j sends this to (z_1, z_2): the composition $T = j \circ s$ is "go to the next side". By Fact 2.6, $T(t) = t + \omega$ with $\omega = -c$.

Let $f(t)$ be the first coordinate φ of the point t. Since j does not change φ, $f(-t) = f(t)$; f has degree 2, and the description of its critical points is the paragraph on even functions of degree 2 in §2.3. The second coordinate of the point t is the first coordinate of $T(t)$, i.e. $f(t+\omega)$. Finally, the polygon through every vertex closes after n steps iff $f(t+n\omega) = f(t)$ for all t; since the fibres of f are the pairs $\{\pm t\}$, this means $t + n\omega \equiv \pm t$ for all t, i.e. $n\omega \in \Lambda$ (the sign $-$ would give $2t \equiv -n\omega$ for all t). The vertices $f(t + j\omega)$, $0 \le j < n$, are then distinct for generic t iff no $k\omega$ with $0 < k < n$ lies in Λ, i.e. iff ω has order n. □

Remark 5 (The degenerate case). Δ is stable under $\varphi \mapsto 1/\bar{\varphi}$ (Remark 2) and has no root on Γ, so its roots are $r, 1/\bar{r}, s, 1/\bar{s}$ with $r, s \in \mathbb{D}$. The only way for them not to be distinct is $r = s$: then $\sqrt{\Delta}$ is a polynomial, the two roots $\psi = \big(-Q \pm \sqrt{\Delta}\big)/(2P)$ are rational functions of φ, and F factors into the two graphs $\psi = h_1(\varphi)$, $\psi = h_2(\varphi)$, of bidegrees $(1, d_1)$ and $(1, d_2)$ with $d_1 + d_2 = 2$. The symmetry of F maps a curve of bidegree $(1, d)$ to one of bidegree $(d, 1)$, which must be one of the two

components; this forces $d_1 = d_2 = 1$, so h_1, h_2 are Möbius maps. The symmetry $s\colon (\varphi, \psi) \mapsto (\psi, \varphi)$ permutes them. If it preserved the graph of h_1, then h_1 would be an involution, without fixed points on Γ (since $h_1(z) \neq z$); it would then preserve the orientation of Γ (an orientation-reversing homeomorphism of a circle has fixed points), hence $\mathbb{D}$, and be of the form $z \mapsto \frac{P-z}{1-\bar{P}z}$, and all the chords $[z, h_1(z)]$ would pass through P (computation in the proof of Lemma 4, case $k = n/2$, below), which is impossible for tangents to an ellipse. So s exchanges the two graphs, i.e. $h_2 = h_1^{-1}$: with $M = h_1$, $\mathcal{C}$ is the union of the graphs of M and M^{-1}, and the vertices of the Poncelet polygons are the orbits of M. When the polygons close, this is the situation of Proposition 1 below (Remark 6): after a projective transformation preserving the disk, γ becomes a circle concentric with Γ.

Proposition 1 (Elliptic disk automorphisms; [8]). *Let $h \in \mathrm{Aut}(\mathbb{D})$, $R(z) = e^{i\theta}z$ with $\theta \in (0, 2\pi)$, $\theta \neq \pi$, and $M = h^{-1} \circ R \circ h$.*

1. *All the chords $[z, M(z)]$, $z \in \Gamma$, are tangent to one ellipse $\gamma \subset \mathbb{D}$, whose foci are $M(0)$ and $M^{-1}(0)$.*
2. *If $\theta = 2\pi/n$ with $n \geq 3$, the points $z, M(z), \ldots, M^{\circ(n-1)}(z)$ are, in counterclockwise order, the vertices of a polygon circumscribing γ, and the Blaschke product $B(z) = z\,M(z)\,M^{\circ 2}(z) \cdots M^{\circ(n-1)}(z)$ is associated to γ.*

Proof. Use the *Klein model* of the hyperbolic plane: the restrictions to Γ of the automorphisms of $\mathbb{D}$ are exactly the restrictions to Γ of the projective transformations of the plane which preserve the disk. Such a transformation maps lines to lines and conics to conics. Let H be the one which extends h^{-1}. The chords $[w, R(w)]$, $w \in \Gamma$, are at distance $|\cos(\theta/2)| > 0$ from 0 (as $\theta \neq \pi$), i.e. tangent to the circle $|w| = |\cos(\theta/2)|$. Since $M = h^{-1} \circ R \circ h$, the chords $[z, M(z)]$ are the images by H of the chords $[w, R(w)]$, hence tangent to the image of this circle by H: a conic contained in $\mathbb{D}$ (because H preserves $\mathbb{D}$), i.e. an ellipse γ. Through $z_0 \in \Gamma$ pass the two tangents $[z_0, M(z_0)]$ and $[M^{-1}(z_0), z_0]$, so by Lemma 3, $M(z_0)M^{-1}(z_0) = \frac{(z_0-\alpha)(z_0-\beta)}{(1-\bar{\alpha}z_0)(1-\bar{\beta}z_0)}$ for all $z_0 \in \Gamma$. Two rational functions equal on Γ are equal, and the zeros of $M(z)M^{-1}(z)$ are $M^{-1}(0)$ and $M(0)$: these are the foci.

For item 2: the orbit of R is a regular n-gon whose consecutive vertices are w and $R(w)$, and h^{-1} preserves the circular order on Γ, so consecutive vertices of the orbit of M are z and $M(z)$. Next, $B(M(z)) = M(z) \cdots M^{\circ n}(z) = B(z)$, so B is constant on the orbits of M; since B has degree n, its fibres on Γ are exactly these orbits. □

In this situation F is reducible (it is the union of the graphs of M and M^{-1}), and B has a single critical value in $\mathbb{D}$: this is case (a) of Theorem Z. Conversely:

Remark 6 (The degenerate case, continued). Assume Δ has a double root and the polygons circumscribing γ are n-gons. By Remark 5, their vertices are the orbits of a Möbius map M; replacing M by M^{-1} if necessary, M sends each $z \in \Gamma$ to its counterclockwise neighbour. Thus M maps Γ to itself without fixed point, and $M^{\circ n} = \mathrm{id}$ on Γ, hence everywhere. A homeomorphism of the circle reversing the orientation has fixed points, so M preserves the orientation of Γ, hence maps $\mathbb{D}$ onto $\mathbb{D}$: $M \in \mathrm{Aut}(\mathbb{D})$. An automorphism of finite order fixes a point of $\mathbb{D}$ (the hyperbolic centre of any orbit), so $M = h^{-1} \circ R \circ h$ with $h \in \mathrm{Aut}(\mathbb{D})$ and R a rotation; since M sends each vertex to the next one in circular order, R is the rotation by $2\pi/n$.

If moreover B is associated to γ, write h^n for the function $z \mapsto h(z)^n$ (a power, not an iterate). Since $h(M(z))^n = \left(e^{2i\pi/n}h(z)\right)^n = h(z)^n$, the function h^n is constant on the orbits of M, so B and h^n have the same fibres on Γ, and as in the proof of Theorem X, $B = \mu \circ h^n$ with μ of degree 1, mapping $\mathbb{D}$ onto $\mathbb{D}$: $\mu \in \mathrm{Aut}(\mathbb{D})$. The only critical point of h^n in $\mathbb{D}$ is $h^{-1}(0)$ (of multiplicity $n-1$), so B has a single critical value in $\mathbb{D}$: case (a) of Theorem Z. (Compare [1, Thm. 4.7], which gives this normal form for any Blaschke product with one critical value.)

4.1 Proof of Lemma 4

We first need the converse of Lemma 1.

Lemma 6 (Converse of Lemma 1)**.** *Every symmetric polynomial $G(\varphi,\psi)$ of bidegree $(2,2)$ can be written $G = \mathbf{d}(\varphi,\psi)^T M\,\mathbf{d}(\varphi,\psi)$ for a unique complex symmetric 3×3 matrix M. In words:* every symmetric $(2,2)$ relation between two points of the circle is the relation "the chord is tangent to a fixed conic" *(possibly complex or degenerate).*

Proof. The coordinates of $\mathbf{d}$ span $\langle 1, \varphi\psi, \varphi+\psi\rangle$, the space of symmetric forms of bidegree $(1,1)$. Their six pairwise products $1,\ \varphi+\psi,\ \varphi\psi,\ (\varphi+\psi)^2,\ \varphi\psi(\varphi+\psi),\ \varphi^2\psi^2$ span the 6-dimensional space of symmetric forms of bidegree $(2,2)$, since $\varphi^2+\psi^2 = (\varphi+\psi)^2 - 2\varphi\psi$. Uniqueness follows by counting dimensions. □

Proof of Lemma 4, generic case. Use Lemma 5: the vertices of a polygon are $z_i = f(t+i\omega)$, so the sides of P_k are the chords $\big(f(t), f(t+k\omega)\big)$. Let $\mathcal{C}_k$ be the image of $\Psi_k\colon t \mapsto \big(f(t), f(t+k\omega)\big)$.

$\mathcal{C}_k$ is a symmetric irreducible curve of bidegree $(2,2)$. Two parameters t, t' have the same image iff $t' = \pm t$ and $t' + k\omega = \pm(t+k\omega)$. Apart from $t' = t$, this forces either $2t \in \Lambda$ (four values of t) or $2k\omega \in \Lambda$. So if $2k\omega \notin \Lambda$, i.e. $2k \not\equiv 0 \pmod n$, Ψ_k is injective outside finitely many points; above a generic $\varphi = f(t)$ lie the two points with $\psi = f(\pm t + k\omega)$, hence bidegree $(2,2)$. The curve is irreducible (image of a torus), and symmetric because $\big(f(t+k\omega), f(t)\big) = \Psi_k(-t-k\omega)$. Its equation F_k is then symmetric (an antisymmetric one would be divisible by $\varphi-\psi$, hence reducible).

A conic. By Lemma 6, $F_k = \mathbf{d}^T M_k \mathbf{d}$. If $\det M_k = 0$, the quadratic form M_k would be a product of two linear forms and F_k would be reducible; so M_k is invertible and the chords $z_i z_{i+k}$ are tangent to the conic γ_k of matrix $A_k = M_k^{-1}$ (§2.7).

The conic is real. The chords for $f(t) \in \Gamma$ are real lines (Remark 2); they lie on the dual conic of M_k and, by conjugation, on that of $\overline{M_k}$. These two irreducible dual conics share infinitely many points, so $\overline{M_k}$ is proportional to M_k by Bézout, and M_k can be chosen real.

It is an ellipse inside $\mathbb{D}$. The map "next vertex" $z \mapsto T(z)$ is a homeomorphism of Γ without fixed point with $T^n = \mathrm{id}$; an orientation-reversing homeomorphism of a circle has fixed points, so T, and T^k, preserve the orientation. Hence when z moves counterclockwise on Γ, both endpoints $P_1 = e^{ia} = z$ and $P_2 = e^{ib} = T^k(z)$ of the chord move counterclockwise: $a' > 0$, $b' > 0$. With the normal $\nu = i(P_2 - P_1)$, the normal velocities of the endpoints are

$$\mathrm{Re}\big(\bar\nu\, P_1'\big) = a'\big(\cos(b-a) - 1\big) < 0, \qquad \mathrm{Re}\big(\bar\nu\, P_2'\big) = b'\big(1 - \cos(b-a)\big) > 0.$$

The normal velocity of the point $(1-u)P_1 + uP_2$ of the moving line is the linear interpolation of these two numbers, so it vanishes for some $u \in (0,1)$. This point, where the line "pivots", is the point where the line touches its envelope, i.e. its point of tangency with γ_k. So γ_k has real points inside $\mathbb{D}$. It does not meet Γ: a point of $\Gamma \cap \gamma_k$ would lie on a single tangent, whereas every point of Γ lies on two distinct chords of P_k. A real conic with a connected component inside $\mathbb{D}$ is an ellipse contained in $\mathbb{D}$ (parabolas and hyperbolas are unbounded). □

Proof of Lemma 4, case $k = n/2$. Now $2k\omega \in \Lambda$, and Ψ_k is 2-to-1 onto its image (since $-t + k\omega \equiv -(t+k\omega)$): the map $f(t) \mapsto f(t+k\omega)$ is well defined, and its graph, the image of Ψ_k, has bidegree $(1,1)$, so it is a Möbius map ι. It is an involution, maps Γ to Γ without fixed point there (the vertices z_i and $z_{i+n/2}$ are distinct), hence preserves the orientation of Γ and maps $\mathbb{D}$ onto $\mathbb{D}$. Such an involution is $\iota(z) = \frac{P-z}{1-\bar P z}$ for a unique $P \in \mathbb{D}$ (it is the unique involution exchanging 0 and $P = \iota(0)$: if ι' is another one, $\iota' \circ \iota$ fixes 0, so it is a rotation, and an involution of the form $\iota' = R_\theta \circ \iota$

with $P \neq 0$ forces $R_\theta(P) = P$, i.e. $\theta = 0$; for $P = 0$ the only fixed-point-free involution fixing 0 is $z \mapsto -z$). The chord $[z, \iota(z)]$ passes through P: for $|z| = 1$,

$$\frac{\iota(z) - P}{z - P} = \frac{-z\,(1 - |P|^2)}{(1 - \bar{P}z)(z - P)} = \frac{-(1 - |P|^2)}{\bar{z}(1 - \bar{P}z)(z - P)} = -\frac{1 - |P|^2}{|z - P|^2} < 0,$$

using $\bar{z}(1 - \bar{P}z) = \bar{z} - \bar{P}$. So P lies on the segment $[z, \iota(z)]$, and all long diagonals pass through $P_{n/2} = P$. □

Proof of Lemma 4, degenerate case. The vertices are the orbits of $M = h^{-1} \circ R \circ h$, with R the rotation by $2\pi/n$ (Remark 6). The sides of P_k are the chords $[z, M^k(z)]$, and $M^k = h^{-1} \circ R^k \circ h$ where R^k is the rotation by $2\pi k/n$. If $2k \not\equiv 0 \pmod{n}$, Proposition 1(1) applies; if $2k \equiv 0$, M^k is a fixed-point-free involution and the argument of the case $k = n/2$ applies. □

5 Theorem Z: characterization through critical values

Theorem Z. *Let B be a Blaschke product of degree $n \geq 3$. Then B is associated to an ellipse if and only if one of the following holds:*

(a) B has exactly one critical value in $\mathbb{D}$;

(b) B has exactly two critical values v_1, v_2 in $\mathbb{D}$, and all its critical points in $\mathbb{D}$ are simple.

The condition is checked without computing any ellipse: find the $n - 1$ roots in $\mathbb{D}$ of $B'(z) = 0$ and compare the values of B there. Case (a) is exceptional (the ellipse is projectively a concentric circle); case (b) is the generic case.

Remark 7. Being associated to an ellipse is invariant under $B \mapsto \mu \circ B$ with $\mu \in \mathrm{Aut}(\mathbb{D})$, which does not change the fibres; so is the condition of Theorem Z. The normalization $B(0) = 0$ plays no role here.

Corollary 1. *Let B be as in case (b), and $C(w) = w\,\dfrac{s - w}{1 - \bar{s}w}$ with $s \in \mathbb{D}$. Then $C \circ B$ (of degree $2n$) is associated to an ellipse if and only if*

$$s = \frac{v_1 + v_2 - v_1 v_2(\bar{v}_1 + \bar{v}_2)}{1 - |v_1 v_2|^2}.$$

5.1 Loops, permutations and the monodromy group

The proof of Theorem Z given in §5.2 is elementary. Freeze one polygon $B^{-1}(\lambda_0)$, let the parameter leave the circle, go around a critical value and come back: the vertices move and come back permuted. The ellipse exists exactly when these permutations never break the polygon, that is, when *neighbors always stay neighbors*; in group-theoretic terms, exactly when the *monodromy group* of B is contained in the dihedral group of the polygon. A count of critical points then shows that this happens exactly in cases (a) and (b), and identifies the monodromy group: cyclic in case (a), dihedral in case (b) (Theorem Z′ below). The elliptic curves of Sections 3 and 4 are not needed; they reappear in §5.4, which explains what lies behind the theorem.

The covering. Let $W \subset \mathbb{P}^1$ be the finite set of critical values of B (possibly including ∞, when B has a multiple pole). By Fact 2.1, $W \cap \Gamma = \emptyset$ and W is symmetric under $w \mapsto 1/\bar{w}$. Outside W, every value has exactly n preimages, and $B' \neq 0$ at each of them, so

$$B\colon\ \mathbb{P}^1 \setminus B^{-1}(W) \ \longrightarrow\ \mathbb{P}^1 \setminus W$$

is a covering map of degree n (it is a local homeomorphism, and it is proper). Consequently, for every path c in $\mathbb{P}^1 \setminus W$ and every point z above its origin, there is a unique *lift* of c starting at z: a path $\tilde{c}$ with $B \circ \tilde{c} = c$ and $\tilde{c}(0) = z$; and two paths in $\mathbb{P}^1 \setminus W$ with the same endpoints which are homotopic (with fixed endpoints) have lifts with the same endpoints (monodromy theorem; see [14]). Concretely, lifting c from z means following the solution of $B(\zeta) = c(s)$ that starts at z as s goes from 0 to 1.

Labels and permutations. Fix $\lambda_0 \in \Gamma$ and label the solutions of $B(z) = \lambda_0$ counterclockwise as $z_0, \dots, z_{n-1}$, indices in $\mathbb{Z}/n$. For a loop ℓ in $\mathbb{P}^1 \setminus W$ based at λ_0, the lift of ℓ starting at z_j ends at a point of $B^{-1}(\lambda_0)$, which we call $z_{\pi_\ell(j)}$. This defines a permutation π_ℓ of $\mathbb{Z}/n$, the *permutation of the loop* ℓ; it only depends on the homotopy class of ℓ in $\mathbb{P}^1 \setminus W$. If $\ell\ell'$ denotes the loop ℓ followed by ℓ', then $\pi_{\ell\ell'} = \pi_{\ell'} \circ \pi_\ell$ (permutations are composed as maps, from right to left), and $\pi_{\ell^{-1}} = \pi_\ell^{-1}$.

Definition 4. The *monodromy group* of B is the subgroup $G_B = \{\pi_\ell\}$ of the symmetric group of $\mathbb{Z}/n$ formed by the permutations of all loops in $\mathbb{P}^1 \setminus W$ based at λ_0.

This is the monodromy group of the branched covering $B\colon \mathbb{P}^1 \to \mathbb{P}^1$ in the usual sense [1, §5]; our labelling of the fibre by $\mathbb{Z}/n$ in counterclockwise order makes it a well-defined subgroup, not only a subgroup up to conjugation. Changing λ_0 to another point of Γ, or shifting the labels, conjugates G_B by a rotation $j \mapsto j + c$, which does not affect the statements below.

Fact 5.1. *1. (*Small loops.*) Let $v \in W$, and let ℓ_v follow a path c_v from λ_0 to a point near v, circle once counterclockwise around v, and come back along c_v. Then the cycles of $\sigma_v := \pi_{\ell_v}$ have as lengths the local degrees $e_B(z)$ of B at the points $z \in B^{-1}(v)$: one cycle of length $e_B(z)$ for each z.*

*2. (*Generators.*) Let $v_1, \dots, v_r$ be the critical values of B in $\mathbb{D}$, so that the critical values outside $\overline{\mathbb{D}}$ are $1/\bar{v}_1, \dots, 1/\bar{v}_r$. The paths c_v can be chosen pairwise disjoint (except at λ_0), inside $\mathbb{D}$ for $v_1, \dots, v_r$ and outside $\overline{\mathbb{D}}$ for the others, in such a way that, after renumbering the v_i,*

- *the circle Γ, traversed once counterclockwise from λ_0, is homotopic in $\mathbb{P}^1 \setminus W$ to the product $\ell_{v_1}\ell_{v_2}\cdots\ell_{v_r}$;*
- *the circle Γ traversed once clockwise is homotopic to a product of the $\ell_{1/\bar{v}_i}$, in some order;*
- *every loop in $\mathbb{P}^1 \setminus W$ based at λ_0 is homotopic to a product of the loops ℓ_v, $v \in W$, and of their inverses.*

In particular G_B is generated by the $2r$ permutations σ_v, $v \in W$.

Why. (1) Let $z \in B^{-1}(v)$ with $e = e_B(z)$. By §2.1, in a suitable local coordinate u centred at z, $B = v + u^e$ (or $1/B = u^e$ if $v = \infty$). If the end point λ' of c_v and the small circle around v lie in the image of this coordinate disc, the solutions of $B(\zeta) = w$ near z are the e roots u of $u^e = w - v$, and when $w - v$ goes once around 0 they are permuted cyclically ($u \mapsto e^{2i\pi/e}u$). Lifting c_v from z_j to a solution near some z, then the small circle, then c_v backwards, shows that the points of $B^{-1}(\lambda_0)$ corresponding to the solutions near z form one cycle of σ_v, of length e.

(2) $\mathbb{P}^1 \setminus W$ is a sphere with $2r$ punctures, r of them in $\mathbb{D}$ and r outside $\overline{\mathbb{D}}$; its fundamental group is generated by lassos around the punctures, and the boundary of a disc containing some of the punctures is homotopic to the product of the lassos around these punctures, in the circular order in which the paths c_v leave the base point (see [14, 13]). Apply this to the discs $\mathbb{D}$ and $\mathbb{P}^1 \setminus \overline{\mathbb{D}}$, whose common boundary is Γ with opposite orientations. □

By Fact 2.1(4), the lift of the loop Γ (counterclockwise) starting at z_j is an arc of Γ going counterclockwise from z_j to z_{j+1}. Hence the permutation of Γ is the *one-step rotation* $r\colon j \mapsto j+1$, and Fact 5.1(2) gives

$$\sigma_{v_r} \circ \cdots \circ \sigma_{v_2} \circ \sigma_{v_1} = r. \tag{5}$$

(The order of the factors will play no role.) In particular G_B contains r, so it acts transitively on $\mathbb{Z}/n$.

The index. For a permutation σ of $\mathbb{Z}/n$, let $\mathrm{ind}(\sigma) = n - (\text{number of cycles of } \sigma)$, fixed points counting as cycles of length 1. By Fact 5.1(1), the number of cycles of σ_v is $\#B^{-1}(v)$, so by (1)

$$\mathrm{ind}(\sigma_v) = \sum_{z \in B^{-1}(v)} \big(e_B(z) - 1\big)$$

is the number of critical points above v, counted with multiplicity. For $v \in \mathbb{D}$ the fibre $B^{-1}(v)$ lies in $\mathbb{D}$, every critical point in $\mathbb{D}$ lies above some v_i, and B has $n-1$ critical points in $\mathbb{D}$ counted with multiplicity (Fact 2.1(2)). Hence

$$\mathrm{ind}(\sigma_{v_1}) + \cdots + \mathrm{ind}(\sigma_{v_r}) = n - 1. \tag{6}$$

Note also that $\sigma_{v_i} \neq \mathrm{id}$: there is a critical point above v_i, hence a cycle of length ≥ 2.

Neighbors and the dihedral group. The *neighbor graph* is the polygon $z_0 z_1 \cdots z_{n-1}$ seen as a graph: vertices $\mathbb{Z}/n$, edges $\{j, j+1\}$. A permutation of $\mathbb{Z}/n$ *preserves neighbors* if it maps edges to edges. The *dihedral group* D_n is the group of order $2n$ formed by the n rotations $j \mapsto j + c$ and the n reflections $j \mapsto c - j$ ($c \in \mathbb{Z}/n$): the symmetries of the regular n-gon with vertices labelled by $\mathbb{Z}/n$.

Lemma 7. *For $n \geq 3$, a permutation of $\mathbb{Z}/n$ preserves neighbors if and only if it belongs to D_n. Consequently the permutations preserving neighbors form a group, D_n.*

Proof. Rotations and reflections clearly preserve neighbors. Conversely, let π preserve neighbors and put $c = \pi(0)$. Since $\{0, 1\}$ is an edge, $\pi(1) = c + \varepsilon$ with $\varepsilon = \pm 1$. We show by induction on $j \geq 1$ that $\pi(j) = c + \varepsilon j$. Suppose $\pi(j-1) = c + \varepsilon(j-1)$ and $\pi(j) = c + \varepsilon j$. Since $n \geq 3$, the vertex $j+1$ is the neighbor of j other than $j-1$; so $\pi(j+1)$ is a neighbor of $\pi(j) = c + \varepsilon j$ other than $\pi(j-1) = c + \varepsilon(j-1)$, that is, $\pi(j+1) = c + \varepsilon(j+1)$. So π is the rotation $j \mapsto j + c$ if $\varepsilon = 1$ and the reflection $j \mapsto c - j$ if $\varepsilon = -1$. □

We also need the index of these permutations.

- A rotation $j \mapsto j + c$ of order $m \geq 2$ (so $m \mid n$) has n/m cycles, all of length m: ind $= n - n/m \geq n/2$.
- A reflection $j \mapsto c - j$ is an involution. Its fixed points are the solutions of $2j \equiv c \pmod{n}$: exactly one if n is odd, none or two if n is even. The other points form 2-cycles. Hence ind $= \frac{n-1}{2}$ if n is odd, and ind $\in \{\frac{n}{2}, \frac{n}{2} - 1\}$ if n is even. In all cases ind $\geq \frac{n}{2} - 1$, and ind $= 1$ when $n = 3$.

From a tangency relation to an ellipse. The last tool turns an algebraic tangency relation into a genuine ellipse. Recall from §2.7 that a line $L = (L_x, L_y, L_c)$ passes through the point $p = (p_x, p_y, p_z)$ of the projective plane iff $L^T p = 0$; thus a linear form in L is the condition "L passes through a fixed point".

Lemma 8. *Let B have degree $n \geq 3$, and suppose there is a complex symmetric 3×3 matrix $M \neq 0$ such that, for every $\lambda \in \Gamma$ and every two consecutive vertices z, z' of the polygon $B^{-1}(\lambda)$, $\mathbf{d}(z,z')^T M \mathbf{d}(z,z') = 0$. Then M is invertible and can be chosen real, and the conic γ of matrix M^{-1} is an ellipse contained in $\mathbb{D}$. In particular, B is associated to γ.*

Proof. Call *side* any segment $[z, z']$ joining two consecutive vertices of a polygon $B^{-1}(\lambda)$, $\lambda \in \Gamma$; every point of Γ is a vertex of exactly one polygon and lies on exactly two sides, which are distinct lines since $n \geq 3$.

M is invertible. Otherwise the quadratic form $L \mapsto L^T M L$ has rank at most 2, so it is a product of two linear forms: $L^T M L = (L^T p)(L^T q)$ for two points P, Q of the complex projective plane (with coordinate vectors p, q). The hypothesis says that every side passes through P or through Q. If $P = Q$, the two sides at a vertex $z \neq P$ both pass through z and P, hence coincide: impossible. So $P \neq Q$; at a vertex $z \notin \{P, Q\}$ the two sides cannot both pass through P (they would both be the line (zP)), nor both through Q: one passes through P and the other through Q. Thus, for every $z \in \Gamma \setminus \{P, Q\}$, the line (zP) is a side: a real line (Remark 2), which meets Γ at z and at a second point, a neighbor of z. We claim that this forces P to be a real point of the open disc $\mathbb{D}$:

- if P is not real, it lies on exactly one real line (the line joining P and $\bar{P}$), which contains at most two points of Γ; but the lines (zP), $z \in \Gamma \setminus \{P, Q\}$, are real, and infinitely many of them are distinct, since a line contains at most two points of Γ;
- if $P \in \Gamma$, then P is the second intersection of (zP) with Γ, hence a neighbor of every $z \in \Gamma \setminus \{P, Q\}$; but P has only two neighbors;
- if P is a real point outside $\overline{\mathbb{D}}$, or a real point at infinity, some line through P is tangent to Γ at a point z, and then (zP) does not meet Γ again.

The same holds for Q. Now for $P \in \mathbb{D}$ the second intersection of the line (zP) with Γ is $\sigma_P(z) = \frac{P - z}{1 - \bar{P} z}$ (computation in the proof of Lemma 4, case $k = n/2$), and $\sigma_P \in \mathrm{Aut}(\mathbb{D})$ is an involution. At most one side lies on the line (PQ), so we may choose λ such that no side of the polygon $B^{-1}(\lambda)$ lies on (PQ); then each side of this polygon passes through exactly one of P, Q, and the two sides at each vertex pass through different points. Going around the polygon, the sides thus alternate: through P, through Q, through P, ... If n is odd this is impossible. If n is even, let $z_0, z_1, \ldots$ be the vertices, with $z_1 = \sigma_P(z_0)$ say; then $z_2 = \sigma_Q(z_1)$, $z_3 = \sigma_P(z_2)$, and so on, so that $z_{2k} = g^k(z_0)$ with $g = \sigma_Q \circ \sigma_P$ (or $z_{2k} = g^{-k}(z_0)$ if the first side passes through Q). Hence $g^{n/2}(z_0) = z_0$ for all but finitely many $z_0 \in \Gamma$ (those of the excluded polygon), so $g^{n/2} = \mathrm{id}$: g has finite order. But $g \neq \mathrm{id}$, since $\sigma_P(0) = P \neq Q = \sigma_Q(0)$ gives $\sigma_Q \neq \sigma_P$. And in the matrices $\sigma_P \sim \begin{pmatrix} -1 & P \\ -\bar{P} & 1 \end{pmatrix}$, the product g has trace $t = 2 - 2\,\mathrm{Re}(\bar{P}Q)$ and determinant $\delta = (1 - |P|^2)(1 - |Q|^2) > 0$. Since $t/2 \geq 1 - |P||Q| > 0$,

$$\tfrac{t^2}{4} - \delta \;\geq\; (1 - |P||Q|)^2 - (1 - |P|^2)(1 - |Q|^2) = (|P| - |Q|)^2 \;\geq 0,$$

i.e. $t^2 \geq 4\delta$: g is parabolic or hyperbolic (not elliptic), and such a Möbius map, other than the identity, has infinite order. This contradiction shows $\det M \neq 0$.

The conic is real. By Remark 2, the sides are real lines, i.e. L is proportional to $\overline{L}$; so they also satisfy $L^T \overline{M} L = 0$. The dual conics of M and $\overline{M}$ are non-degenerate ($\det \overline{M} = \overline{\det M} \neq 0$) and

share infinitely many points (the sides), so by Bézout (§2.7) they coincide: $\overline{M}$ is a non-zero multiple of M, say $\overline{M} = \kappa M$ with $|\kappa| = 1$ ($\kappa\bar{\kappa}M = \overline{\overline{M}} = M$), and $\kappa^{-1/2}M$ is real. So after rescaling M is real, and the conic γ of matrix M^{-1} is real, non-degenerate, and tangent to all the sides (§2.7).

It is an ellipse inside $\mathbb{D}$. As λ turns counterclockwise on Γ, both endpoints $P_1 = e^{ia}$ and $P_2 = e^{ib}$ of a side move counterclockwise with positive speed (Fact 2.1(4); $B' \neq 0$ on Γ, so the lifts are smooth): $a' > 0$, $b' > 0$. With the normal $\nu = i(P_2 - P_1)$, the normal velocities of the endpoints are

$$\mathrm{Re}\big(\bar{\nu}\, P_1'\big) = a'\big(\cos(b-a) - 1\big) < 0, \qquad \mathrm{Re}\big(\bar{\nu}\, P_2'\big) = b'\big(1 - \cos(b-a)\big) > 0$$

(for the first one, $\bar{\nu}P_1' = -i(e^{-ib} - e^{-ia})\, ia'e^{ia} = a'(e^{i(a-b)} - 1)$; the second is similar). The normal velocity of the point $(1-u)P_1 + uP_2$ of the moving line interpolates linearly between these two numbers, so it vanishes for some $u \in (0,1)$: a point strictly inside the chord, hence in $\mathbb{D}$. There the moving line touches its envelope, and the envelope of a family of tangent lines to the non-degenerate conic γ is γ itself. So γ has real points inside $\mathbb{D}$. It does not meet Γ: through a point of γ passes a single tangent to γ, while through each point of Γ pass two distinct sides, both tangent to γ. A connected component of a real non-degenerate conic which contains a point of $\mathbb{D}$ and does not meet Γ is contained in $\mathbb{D}$, hence bounded; the components of parabolas and hyperbolas are unbounded, so γ is an ellipse, and it lies in $\mathbb{D}$. □

5.2 Proof of Theorem Z

We prove the following more precise statement, which contains Theorem Z.

Theorem Z′ (Monodromy form)**.** *Let B be a Blaschke product of degree $n \geq 3$, and G_B its monodromy group, with respect to a counterclockwise labelling of a fibre $B^{-1}(\lambda_0)$, $\lambda_0 \in \Gamma$ (Definition 4). The following are equivalent:*

1. *B is associated to an ellipse;*
2. *G_B preserves neighbors, i.e. $G_B \subset D_n$;*
3. *B satisfies (a) or (b) of Theorem Z.*

When they hold, G_B is the cyclic group $\langle r \rangle$ of order n in case (a), and the full dihedral group D_n of order $2n$ in case (b).

The proof goes around the following circle of implications:

$$\begin{aligned} \text{associated} &\overset{\text{Step 1}}{\Longrightarrow} G_B \subset D_n \overset{\text{Step 2}}{\Longrightarrow} \text{(a) or (b)} \\ &\overset{\text{Step 3}}{\Longrightarrow} G_B \subset D_n \overset{\text{Step 4}}{\Longrightarrow} \text{associated,} \end{aligned}$$

and Step 3 also identifies G_B. Throughout, $z_0, \ldots, z_{n-1}$ is the fixed labelled fibre over λ_0.

Step 1. An ellipse forces every loop to preserve neighbors. Let B be associated to an ellipse $\gamma \subset \mathbb{D}$, and let $F(\varphi, \psi) = P(\varphi)\psi^2 + Q(\varphi)\psi + R(\varphi)$ be the polynomial of Lemma 1 for γ, with discriminant $\Delta = Q^2 - 4PR$.

(i) On the circle, F finds the neighbors. Let $z \in \Gamma$. By Remark 3, $P(z) \neq 0$ and $\Delta(z) \neq 0$, so $F(z, \cdot)$ has two distinct roots, and these are the second intersections with Γ of the two tangents to γ through z. Since B is associated to γ, these two tangents are the two sides at z of the polygon $B^{-1}(B(z))$. So *the two roots of $F(z, \cdot)$ are the two neighbors of z in its polygon*; in particular both satisfy $B(\psi) = B(z)$.

(ii) This stays true off the circle. Let $E \subset \mathbb{P}^1$ be the finite set consisting of ∞ and of the zeros of P and of Δ; by (i), $E \cap \Gamma = \emptyset$. On $\mathbb{P}^1 \setminus E$, the two roots of $F(z,\cdot)$ are the two branches of

$$\psi_\pm(z) = \frac{-Q(z) \pm \sqrt{\Delta(z)}}{2P(z)},$$

which are distinct, analytic near every point of $\mathbb{P}^1 \setminus E$, and can be continued analytically along every path in $\mathbb{P}^1 \setminus E$ (the only obstruction, a zero of Δ, is in E). Let $z \notin E$ and let ψ be one of the two root germs at z. Choose a path in $\mathbb{P}^1 \setminus E$ from z to a point $u \in \Gamma$ (possible since E is finite) and continue ψ along it: it arrives at u as one of the two root germs there, which, by (i), satisfies $B(\psi(\zeta)) = B(\zeta)$ for all ζ on an arc of Γ around u. Both sides are analytic near u, so by the identity theorem $B(\psi(\zeta)) = B(\zeta)$ on a neighbourhood of u; and an identity between analytic germs is preserved by analytic continuation along the path back to z. Therefore, *for every $z \notin E$, both roots of $F(z,\cdot)$ are solutions of $B(\cdot) = B(z)$.*

(iii) Edges are carried along. Let ℓ be a loop in $\mathbb{P}^1 \setminus W$ based at λ_0. The set $B(E)$ is finite, so, by a small deformation within $\mathbb{P}^1 \setminus W$ (which does not change π_ℓ), we may assume that ℓ avoids $B(E)$. Let $z_k(s)$ be the lift of ℓ starting at z_k ($k \in \mathbb{Z}/n$, $0 \le s \le 1$). Since ℓ avoids W, the n points $z_0(s), \dots, z_{n-1}(s)$ are pairwise distinct for every s; since ℓ avoids $B(E)$, none of them lies in E. Fix j. Along the path $s \mapsto z_j(s)$, which stays in $\mathbb{P}^1 \setminus E$, the two roots of $F(z_j(s),\cdot)$ are given by the analytic continuation of $\psi_\pm$ and depend continuously on s; by (ii) they lie in the finite set $\{z_0(s), \dots, z_{n-1}(s)\}$, whose points depend continuously on s and stay pairwise distinct. A continuous choice among finitely many continuously moving, pairwise distinct points is locally constant, hence constant: if a root equals $z_k(s)$ for one s, it equals $z_k(s)$ for all s. At $s = 0$ the two roots are, by (i), the two neighbors z_{j-1} and z_{j+1} of z_j. Following them to $s = 1$, where $z_k(1) = z_{\pi_\ell(k)}$, we find that the two roots of $F(z_{\pi_\ell(j)},\cdot)$ are $z_{\pi_\ell(j-1)}$ and $z_{\pi_\ell(j+1)}$. But by (i) these two roots are $z_{\pi_\ell(j)-1}$ and $z_{\pi_\ell(j)+1}$. Hence

$$\{\pi_\ell(j-1), \pi_\ell(j+1)\} = \{\pi_\ell(j) - 1, \pi_\ell(j) + 1\} \qquad \text{for all } j,$$

i.e. π_ℓ maps edges to edges. By Lemma 7, $\pi_\ell \in D_n$ for every loop ℓ: $G_B \subset D_n$.

Step 2. If $G_B \subset D_n$, then (a) or (b). Let $v_1, \dots, v_r$ be the critical values in $\mathbb{D}$ and $\sigma_i = \sigma_{v_i} \in G_B$ the permutations of the small loops. By assumption each σ_i is a rotation or a reflection, and $\sigma_i \neq \mathrm{id}$ (there is a critical point above v_i). Let ρ be the number of rotations and ϕ the number of reflections among $\sigma_1, \dots, \sigma_r$, so $\rho + \phi = r$. Reflections are exactly the elements of D_n which reverse the cyclic orientation of $\mathbb{Z}/n$ (they are the elements $j \mapsto c - j$), so the map "orientation" $D_n \to \{\pm 1\}$ is a homomorphism, equal to -1 exactly on reflections. By (5) the product of the σ_i is the rotation r, of orientation $+1$; hence ϕ is even. Now (6) says that the indices of the σ_i add up to exactly $n-1$, while each rotation (of order ≥ 2) has index $\ge n/2$ and each reflection has index $\ge n/2 - 1$:

- if $\rho \ge 2$: total index $\ge n > n-1$, impossible;
- if $\rho = 1$ and $\phi \ge 2$: total index $\ge \frac{n}{2} + 2(\frac{n}{2} - 1) = \frac{3n}{2} - 2 > n - 1$ since $n > 2$, impossible;
- if $\rho = 0$ and $\phi \ge 4$: total index $\ge 4(\frac{n}{2} - 1) = 2n - 4 > n - 1$ for $n \ge 4$, and $\ge 4 \cdot 1 > 2$ for $n = 3$, impossible;
- if $\rho = \phi = 0$: no critical value in $\mathbb{D}$, impossible since B has $n - 1 \ge 2$ critical points in $\mathbb{D}$.

Only two possibilities remain. Either $(\rho, \phi) = (1, 0)$: a single critical value in $\mathbb{D}$, case (a). Or $(\rho, \phi) = (0, 2)$: exactly two critical values v_1, v_2 in $\mathbb{D}$, and σ_1, σ_2 are reflections, hence involutions. An involution has only cycles of length 1 and 2, so by Fact 5.1(1) every critical point above v_1 or v_2, that is, every critical point in $\mathbb{D}$, has local degree 2: case (b).

Step 3. In cases (a) and (b), $G_B \subset D_n$; identification of G_B. By Fact 5.1(2), G_B is generated by the permutations σ_v of the small loops around all critical values, in $\mathbb{D}$ and outside $\overline{\mathbb{D}}$. Since D_n is a group (Lemma 7), it suffices to show that each σ_v lies in D_n.

Case (a). There is one critical value v_1 in $\mathbb{D}$ and one, $1/\bar{v}_1$, outside $\overline{\mathbb{D}}$. Equation (5) reads $\sigma_{v_1} = r$, and the clockwise circle gives $\sigma_{1/\bar{v}_1} = r^{-1}$. So $G_B = \langle r \rangle$ is cyclic of order n.

Case (b). The critical values are v_1, v_2 in $\mathbb{D}$ and $1/\bar{v}_1, 1/\bar{v}_2$ outside. All critical points of B are simple: those in $\mathbb{D}$ by hypothesis, those outside $\overline{\mathbb{D}}$ because they are the images of the former by $z \mapsto 1/\bar{z}$, which preserves local degrees (Fact 2.1(1)). So by Fact 5.1(1) each σ_v has only cycles of length 1 and 2: it is an involution, $\sigma_v^{-1} = \sigma_v$. Write $\sigma_1 = \sigma_{v_1}$, $\sigma_2 = \sigma_{v_2}$; by (5), $\sigma_2\sigma_1 = r$ (or $\sigma_1\sigma_2 = r$; exchange the names if necessary). Then

$$\sigma_1\, r\, \sigma_1 = \sigma_1\sigma_2\sigma_1\sigma_1 = \sigma_1\sigma_2 = \sigma_1^{-1}\sigma_2^{-1} = (\sigma_2\sigma_1)^{-1} = r^{-1}.$$

Apply both sides to $\sigma_1(j)$: $\sigma_1(j+1) = \sigma_1\big(r(\sigma_1(\sigma_1(j)))\big) = r^{-1}(\sigma_1(j)) = \sigma_1(j) - 1$ for every j. Hence $\sigma_1(j) = \sigma_1(0) - j$: σ_1 is a reflection, and $\sigma_2 = r\sigma_1$, the composition of a reflection and a rotation, is a reflection too. The same computation, with r^{-1} in place of r and the two critical values outside $\overline{\mathbb{D}}$, shows that $\sigma_{1/\bar{v}_1}$ and $\sigma_{1/\bar{v}_2}$ are reflections. So $G_B \subset D_n$. Moreover G_B contains the rotation r of order n and the reflection σ_1, and these two generate D_n (the rotations are the r^k, and the reflections are the $r^k\sigma_1$). So $G_B = D_n$.

Step 4. If $G_B \subset D_n$, then B is associated to an ellipse. Let $U = \{z \in \mathbb{P}^1 :\ B(z) \notin W\} = \mathbb{P}^1 \setminus B^{-1}(W)$; its complement is finite.

(i) Every point of U has two well-defined neighbors. Let $z \in U$ and let c be a path in $\mathbb{P}^1 \setminus W$ from λ_0 to $B(z)$. The lifts of c starting at $z_0, \dots, z_{n-1}$ end at the n distinct points of $B^{-1}(B(z))$, which we label accordingly: z receives the label j if it is the end of the lift starting at z_j. The *neighbors of z along c* are the two points of $B^{-1}(B(z))$ labelled $j-1$ and $j+1$; they are distinct since $n \geq 3$. If c' is another path from λ_0 to $B(z)$ in $\mathbb{P}^1 \setminus W$, then $\ell = c'c^{-1}$ is a loop at λ_0, and the labelling along c' is the labelling along c composed with π_ℓ: the point labelled k along c' is the point labelled $\pi_\ell(k)$ along c (lift c' from z_k, then c^{-1}: one arrives at $z_{\pi_\ell(k)}$, and lifting c back from there returns to the same point). Since π_ℓ preserves neighbors, $\{\pi_\ell(j-1), \pi_\ell(j+1)\} = \{\pi_\ell(j) \pm 1\}$, so the pair of neighbors of z is the same along c and along c'. We write $N(z) = \{\psi_1(z), \psi_2(z)\}$ for this pair. The relation "ψ is a neighbor of z" is symmetric (both are defined by the same labelling of the fibre $B^{-1}(B(z)) = B^{-1}(B(\psi))$), and $N(z) \subset U$.

(ii) Two rational functions. The functions $s = \psi_1 + \psi_2$ and $p = \psi_1\psi_2$ are well defined on U (they do not depend on the order of the two neighbors). They are meromorphic on U: near a point $z^* \in U$, the labelled fibre over $B(z)$ can be obtained by lifting a fixed path to $B(z^*)$ followed by a short path from $B(z^*)$ to $B(z)$, so the neighbors $\psi_1(z), \psi_2(z)$ are, near z^*, analytic functions of z (local inverses of the covering B), possibly with a pole where a neighbor is ∞. At each of the finitely many points $z^* \notin U$ (and at $z^* = \infty$, in the coordinate $1/z$), s and p are analytic on a punctured neighbourhood and have at most a pole there: indeed the neighbors of z are roots in ζ of the polynomial $N(\zeta)D(z) - N(z)D(\zeta)$, where $B = N/D$, whose coefficients are polynomials in z not all vanishing at z^*; by Cauchy's bound, the roots of a polynomial $a_m\zeta^m + \cdots + a_0$ with $a_m \neq 0$ satisfy $|\zeta| \leq 1 + \max_k |a_k/a_m|$, so $|\psi_i(z)| \leq C|z - z^*|^{-N}$ for some C, N, and a function analytic on a punctured neighbourhood of z^* and bounded by a power of $1/|z - z^*|$ has at most a pole at z^* (Riemann). A function meromorphic on the whole sphere is rational: *s and p are rational functions of z.*

(iii) A symmetric relation of bidegree $(2,2)$. For $z \in U$ (with finite neighbors), $N(z)$ is the set of roots of $\psi^2 - s(z)\psi + p(z)$. Multiplying by a common denominator and dividing by the greatest

common divisor of the resulting coefficients, we get a polynomial

$$G(z,\psi) = P(z)\psi^2 - S(z)\psi + R(z) \in \mathbb{C}[z,\psi], \qquad P \neq 0, \quad \gcd(P,S,R) = 1,$$

and a finite set $Z \subset \mathbb{P}^1$ (the points outside U, the zeros of P, the points where a neighbor is ∞, and ∞) such that, for $z \notin Z$, the roots of $G(z,\cdot)$ are exactly the two distinct neighbors of z. We record three properties of G.

- G has no factor depending on z alone ($\gcd(P,S,R) = 1$), and no factor $h(\psi)$ depending on ψ alone: otherwise a root ψ_1 of h would satisfy $G(z,\psi_1) = 0$ for all z, so ψ_1 would be a neighbor of every $z \notin Z$, whereas (by symmetry) the points having ψ_1 as a neighbor are the two neighbors of ψ_1.

- G is square-free: a repeated factor would have degree ≥ 1 in ψ (by the previous item), so $G(z,\cdot)$ would have a double root for every z, whereas its two roots are distinct for $z \notin Z$.

- G is symmetric. Put $\widetilde{G}(z,\psi) = G(\psi,z)$. For $z,\psi \notin Z$,

  $$G(z,\psi) = 0 \iff \psi \in N(z) \iff z \in N(\psi) \iff \widetilde{G}(z,\psi) = 0,$$

  so the zero sets of G and $\widetilde{G}$ coincide outside the finitely many lines $z \in Z$, $\psi \in Z$. No irreducible factor of G is such a line (first item), so every irreducible component of $\{G = 0\}$ has infinitely many points in $\{\widetilde{G} = 0\}$, hence is contained in it; by the Nullstellensatz, each irreducible factor of G divides $\widetilde{G}$, and since G is square-free, G divides $\widetilde{G}$. By symmetry $\widetilde{G}$ divides G, so $\widetilde{G} = cG$ with $c \in \mathbb{C}^*$; applying this twice gives $c^2 = 1$. If $c = -1$, then $G(z,z) = -G(z,z)$, so $G(z,z) = 0$ for all z, and every $z \notin Z$ would be its own neighbor: impossible. So $c = 1$ and G is symmetric.

Since G is symmetric and of degree 2 in ψ, it has degree 2 in z as well: G is a symmetric polynomial of bidegree $(2,2)$. Finally, since the neighbors depend continuously on $z \in U$ and G is continuous, the relation $G(z,\psi) = 0$ for $\psi \in N(z)$ extends by continuity from $U \setminus Z$ to all of U (with finite neighbors).

(iv) The ellipse. By Lemma 6, $G = \mathbf{d}^T M \mathbf{d}$ for a symmetric matrix $M \neq 0$ (as $G \neq 0$). Let $\lambda \in \Gamma$ and let c be the arc of Γ from λ_0 to λ, counterclockwise. By Fact 2.1(4), the lifts of c starting at $z_0, \ldots, z_{n-1}$ are arcs of Γ traversed counterclockwise, which never meet; so they keep their counterclockwise order, and the labels along c of the vertices of $B^{-1}(\lambda)$ are $0, 1, \ldots, n-1$ in counterclockwise order. Hence two consecutive vertices z, z' of $B^{-1}(\lambda)$ are neighbors in the sense of (i), and $\mathbf{d}(z,z')^T M \mathbf{d}(z,z') = G(z,z') = 0$ ($\Gamma \subset U$, and $z' \neq \infty$). Lemma 8 now shows that M is invertible, real up to a factor, and that B is associated to the ellipse γ of matrix M^{-1}.

This completes the proof of Theorem Z′, hence of Theorem Z. □

Remark 8 (Case (a) made explicit). In case (a), Step 3 can be made concrete. Let v be the only critical value in $\mathbb{D}$. By (1), $B^{-1}(v) \cap \mathbb{D}$ is a single point z_0, where B has local degree n. Take $h, \mu \in \mathrm{Aut}(\mathbb{D})$ with $h(z_0) = 0$ and $\mu(0) = v$. Then $\mu^{-1} \circ B \circ h^{-1}$ is a Blaschke product whose only zero is 0, of multiplicity n: it is $e^{i\theta} z^n$. So $B(z) = \mu\big(e^{i\theta} h(z)^n\big)$, the fibres of B on Γ are the orbits of $M = h^{-1} \circ R \circ h$, with R the rotation by $2\pi/n$, and Proposition 1 gives the ellipse and its foci $M(0)$, $M^{-1}(0)$. This direction is due to Aksoy, Arici, Celorrio and Gorkin [1, Thm. 4.7 and Prop. 4.8], using results of [8]. The monodromy group of $z \mapsto z^n$ is visibly cyclic.

Remark 9 (The two monodromy groups). Theorem Z′ says that, among Blaschke products of degree $n \geq 3$, those associated to an ellipse are exactly those whose monodromy group is a group of symmetries of the polygon $B^{-1}(\lambda_0)$: the cyclic group C_n when the ellipse is projectively a concentric

circle, and the dihedral group D_n otherwise. In the generic case (b), the loop around each of the four critical values acts on the polygon as a reflection (an axial symmetry), and the loop Γ as the rotation by one step. In the degree-5 example of §6.1 (case (b)), each of the two small loops in $\mathbb{D}$ therefore acts as a reflection of the pentagon: it exchanges two pairs of vertices and fixes the fifth.

5.3 Proof of Corollary 1

The critical points of $C \circ B$ in $\mathbb{D}$ are the $n-1$ critical points of B, with values $C(v_1)$, $C(v_2)$, and the n preimages under B of the critical point w_C of C, with value $C(w_C)$.

- Case (a) is impossible: w_C is the only preimage of $C(w_C)$, so $C(v_1) = C(v_2) = C(w_C)$ would force $v_1 = v_2 = w_C$.
- All critical points are simple iff $w_C \notin \{v_1, v_2\}$ (otherwise the critical points of B above v_i become of local degree 4).
- Exactly two values occur iff $C(v_1) = C(v_2)$ (by the same uniqueness argument); this already implies $w_C \notin \{v_1, v_2\}$.

Expanding $v_1(s - v_1)(1 - \bar{s}v_2) = v_2(s - v_2)(1 - \bar{s}v_1)$, the terms $|s|^2 v_1 v_2$ cancel, and dividing by $v_1 - v_2$ gives $s + v_1 v_2 \bar{s} = v_1 + v_2$. Conjugating, $\bar{s} = \bar{v}_1 + \bar{v}_2 - \bar{v}_1 \bar{v}_2 s$; substituting back gives $s(1 - |v_1 v_2|^2) = v_1 + v_2 - v_1 v_2(\bar{v}_1 + \bar{v}_2)$. This linear equation in $(s, \bar{s})$ has a unique solution, since $1 - |v_1 v_2|^2 \neq 0$. Finally this solution lies in $\mathbb{D}$ automatically. Let $\sigma \in \mathrm{Aut}(\mathbb{D})$ be the involution exchanging v_1 and v_2 (the half-turn about their hyperbolic midpoint), and $s^* = \sigma(0) \in \mathbb{D}$. The Blaschke product $C^*(w) = w\frac{s^* - w}{1 - \bar{s}^* w}$ has degree 2; the involution exchanging the two points of each of its fibres is an automorphism of $\mathbb{D}$ exchanging 0 and s^* (the fibre of 0), hence is σ (uniqueness, as in the proof of Lemma 4, case $k = n/2$). So $C^*(v_1) = C^*(v_2)$, and by uniqueness of the solution, $s = s^* \in \mathbb{D}$. □

5.4 What lies behind Theorem Z: Blaschke products as shadows of isogenies

The proof above is elementary, but it does not explain *why* two critical values with simple critical points should be the right condition. The elliptic curve of Section 4 does.

Proposition 2. *Let B be associated to γ, and assume Δ has four distinct roots. With the notation of Lemma 5, let $\Lambda' = \Lambda + \mathbb{Z}\omega$, $E' = \mathbb{C}/\Lambda'$, and $\pi\colon E \to E'$ the isogeny of kernel $\{0, \omega, \dots, (n-1)\omega\}$. There is an even elliptic function g of degree 2 on E' such that the following diagram commutes:*

$$\begin{array}{ccc} E & \xrightarrow{\ \pi\ } & E' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ \mathbb{P}^1 & \xrightarrow[B]{} & \mathbb{P}^1 \end{array} \qquad B \circ f = g \circ \pi.$$

How to read this diagram. Going down by f means "take the X-coordinate": a point of the torus becomes a point of the sphere, and t, $-t$ become the same point. Going right by π means "identify the vertices of a polygon". The proposition says that B is what π looks like once we only see X-coordinates: *B is the shadow of an isogeny.*

Proof. The polygons $B^{-1}(\lambda)$ have exactly n distinct vertices $f(t + j\omega)$, so ω has order exactly n (Lemma 5) and the kernel has n elements. Let $\Pi = B \circ f$, an elliptic function on E of degree $2n$.

Since B is associated to γ, $f(t)$ and $f(t+\omega)$ are in the same fibre of B whenever $f(t) \in \Gamma$ (consecutive vertices). This happens on a whole curve of E, so the meromorphic function $\Pi(t+\omega)-\Pi(t)$ vanishes there, hence everywhere. Thus Π is periodic for Λ': it is an elliptic function on E', i.e. $\Pi = g \circ \pi$ for some g on E'. Degrees multiply: $2n = \deg g \cdot n$, so $\deg g = 2$; and g is even because $\Pi(-t) = \Pi(t)$ (f is even) and $\pi(-t) = -\pi(t)$. □

Necessity read off the diagram. The diagram gives a second proof that an associated B satisfies (a) or (b), and shows where the number 2 comes from. If Δ has a double root, we are in the degenerate case and Remark 6 gives case (a). Otherwise, apply the chain rule for local degrees, $e_{h_2 \circ h_1}(p) = e_{h_1}(p)\, e_{h_2}(h_1(p))$, to both sides of $B \circ f = g \circ \pi$. Since π is unramified,

$$e_f(t) \cdot e_B\big(f(t)\big) = e_g\big(\pi(t)\big).$$

By the paragraph on even functions of degree 2, $e_f(t) = 2$ iff $t \in E[2]$ iff $2t \in \Lambda$, and $e_g(\pi(t)) = 2$ iff $\pi(t) \in E'[2]$ iff $2t \in \Lambda'$. Since $\Lambda \subset \Lambda'$, there are three cases:

t	$e_f(t)$	$e_g(\pi(t))$	hence $e_B(f(t))$
$2t \in \Lambda$	2	2	1
$2t \in \Lambda' \setminus \Lambda$	1	2	2
otherwise	1	1	1

Every point of $\mathbb{P}^1$ is $f(t)$ for some t, so we conclude:

- *every critical point of B is simple* (local degree 2, never more);
- *the critical values of B belong to the four-point set $g(E'[2])$*, since $B(f(t)) = g(\pi(t))$ with $\pi(t) \in E'[2]$;
- B has no critical point at the four points $f(E[2])$, i.e. at the points of $\Gamma \cap \gamma$.

Sanity check: $\pi^{-1}(E'[2])$ has $4n$ points, 4 of which are in $E[2]$; the remaining $4n-4$ are identified in pairs by f, giving $2n-2$ critical points, as predicted by Riemann–Hurwitz.

The set of critical values of B is symmetric under $w \mapsto 1/\bar{w}$ and does not meet Γ (Fact 2.1). Having at most 4 elements, it has at most 2 elements in $\mathbb{D}$. If it had only one, v, the $n-1 \geq 2$ critical points of B in $\mathbb{D}$, all simple, would all lie in $B^{-1}(v)$, which by (1) has $n-(n-1)=1$ point: impossible. So we are in case (b).

The converse through the diagram. In case (b) the diagram can be rebuilt from B alone. Let $w_1, \ldots, w_4$ be the four critical values of B (two in $\mathbb{D}$, two outside), $\Omega(w) = \prod_i (w - w_i)$, E' the elliptic curve $y^2 = \Omega(w)$ and X the curve $y^2 = \Omega(B(z))$. The map $(z, y) \mapsto z$ is branched exactly at the non-critical preimages of the w_i; there are four of them, so X has genus 1 by Riemann–Hurwitz, and $(z, y) \mapsto (B(z), y)$ is an isogeny $X \to E'$ of degree n. Following the loop Γ on X shows that its kernel is cyclic, generated by the "next vertex" translation. This gives another proof of (b) $\Rightarrow$ associated, which we do not detail: the elementary proof of §5.2 is shorter.

5.5 Consequences

Remark 10 ($\Gamma \cap \gamma$ seen by B). In the generic case, the four complex points of $\Gamma \cap \gamma$ (the roots of Δ) are exactly the non-critical preimages of the critical values of B: a non-critical point $f(t)$ above a critical value has $\pi(t) \in E'[2]$, i.e. $2t \in \Lambda'$, and is not in the second line of the table, so it is in the

first one: $t \in E[2]$. Two of these points lie in $\mathbb{D}$; they are distributed $(1,1)$ over (v_1, v_2) if n is odd, and $(2,0)$ if n is even. Indeed, if k_i (simple) critical points lie above v_i, then by (1) v_i has $n - k_i$ preimages, of which $n - 2k_i$ are non-critical; $k_1 + k_2 = n - 1$ gives $(n - 2k_1) + (n - 2k_2) = 2$, and $n - 2k_i$ has the parity of n.

Remark 11 (Decompositions). Composing B with a Blaschke product of degree 2, as in Corollary 1, corresponds to composing the isogeny π with an isogeny of degree 2; the condition $C(v_1) = C(v_2)$ says that the two critical values of $\mathbb{D}$ merge. More generally, the kernel K of π is cyclic of order n (it is generated by ω), so it has exactly one subgroup K_m of each order $m \mid n$, and π factors as $E \to E/K_m \to E/K$. Passing to X-coordinates, this factorization of π is consistent with, and suggests another proof of, the theorem of Aksoy, Arici, Celorrio and Gorkin [1, Thm. 3.5]: a Blaschke product of degree n with an elliptical Blaschke curve is a composition $C \circ D$ with $\deg C = k$ for every factor $k > 1$ of n. (To turn this into a proof, one must still check that the two maps induced on the spheres are Blaschke products for suitable coordinates; we do not pursue this here.)

5.6 Comparison with known results

Remark 12 (Number of critical values). For a composition $B_1 \circ \cdots \circ B_r$ with $\deg B_j = k_j$, Aksoy, Arici, Celorrio and Gorkin show that there are at most $\sum_j (k_j - 1)$ distinct critical values [1, Prop. 4.3]; in particular, a Blaschke product of degree 2^r with an elliptical Blaschke curve has at most r critical values [1, Cor. 4.4]. Theorem Z sharpens this to *at most two critical values in $\mathbb{D}$, in every degree.*

Remark 13 (Monodromy). By Theorem Z$'$, the monodromy group of a Blaschke product associated to an ellipse is the cyclic group of order n in case (a) and the dihedral group of order $2n$ in case (b). For $n = 4$ this agrees with the dihedral group of order 8 found in [1, Ex. 5.4]. For compositions of $r \geq 3$ Blaschke products of degree 2 with r distinct critical values, the monodromy group is the much larger iterated wreath product $\mathbb{Z}_2 \wr \cdots \wr \mathbb{Z}_2$ [1, Thm. 5.7]; by Theorem Z, such compositions never have an elliptical Blaschke curve.

Remark 14 (The simplicity condition is necessary). Aksoy, Arici, Celorrio and Gorkin give the example $B(z) = z^4 \frac{z^4 - a}{1 - az^4}$, $0 < a < 1$, a composition of three Blaschke products of degree 2 whose Blaschke curve is not an ellipse [1, Ex. 3.7]. In the light of Theorem Z: writing $B = B_2 \circ (z \mapsto z^4)$ with $B_2(w) = w\frac{w - a}{1 - aw}$, the critical points of B in $\mathbb{D}$ are 0, of local degree 4 (value 0), and the four fourth roots of the critical point of B_2 (common value $B_2(w_c)$). So B has exactly two critical values in $\mathbb{D}$, but a critical point which is not simple: it fails condition (b), as it must.

Remark 15. To the best of our knowledge, the necessity of (a) or (b), and the sufficiency of (b), are new; the sufficiency of (a) is [1, Prop. 4.8].

6 Examples

This section presents three Blaschke products associated to ellipses: one of prime degree 5, which is not a composition and is constructed directly from the equation of an ellipse through its elliptic coordinate, one of degree 4, and one of degree 8 obtained from it by Corollary 1.

How the numbers are obtained. All numbers, tables and figures of this section are produced by the script `examples_blaschke.py` (run `python3 examples_blaschke.py examples` to regenerate the folder `examples/`). For each Blaschke product B, the script:

1. computes the fibres $B^{-1}(\lambda)$ for 40 values of $\lambda \in \Gamma$, sorts each fibre by argument, and forms the sides of the polygons;

2. looks for a dual conic M with $L^T M L = 0$ for all these sides L (a linear least-squares problem in the six entries of M; §2.7). The *residual* is the ratio of the smallest to the largest singular value of this problem: it is of the order of 10^{-16} (rounding errors) when all the sides are tangent to one conic, and much larger otherwise (for instance $6.8 \cdot 10^{-3}$ for the control of §6.3);

3. computes the ellipse $\gamma = \{x^T M^{-1} x = 0\}$ (centre, semi-axes, inclination of the major axis), its foci (the roots of $\mathbf{d}(\varphi, 0)^T M \mathbf{d}(\varphi, 0)$, Lemma 2), and in the same way the ellipses γ_k of the Poncelet package (Lemma 4) and, for even n, the point $P_{n/2}$;

4. computes the critical points of B in $\mathbb{D}$ (roots of $N'D - ND'$ for $B = N/D$) and the critical values, and reports the case of Theorem Z;

5. checks Theorem X: the zeros of B are 0, the foci of the γ_k, and $P_{n/2}$ if n is even.

How to read the figures. The black circle is Γ. Three fibres $B^{-1}(\lambda)$, for $\lambda = e^{2i\pi\theta}$ with $\theta = 0.05, 0.38, 0.71$, are drawn as polygons (dots: the vertices). The thick red curve is the ellipse γ; the dashed curves are the ellipses γ_k enveloping the diagonals $z_j z_{j+k}$. Stars are the zeros of B, crosses its critical points.

6.1 A Blaschke product of degree 5, from the elliptic function

Degree 5 is prime, so no composition argument applies. We construct the example directly from the equation of an ellipse, following Sections 3 and 4 step by step. Nothing in the construction is specific to this ellipse or to $n = 5$: the same recipe works for every shape of ellipse and every $n \geq 3$.

Step 1: an ellipse, up to scale. We fix the shape of γ arbitrarily: centre $0.12 - 0.18\,i$, ratio of the axes $q/p = 0.6$, major axis inclined by 30°. Only the size is left free: $p = \sigma$, $q = 0.6\,\sigma$. This gives the matrix $A(\sigma)$ of γ.

Step 2: the polynomials of Lemma 1. From the entries $a, \ldots, f$ of $A(\sigma)$ we compute $a_1, \ldots, a_6$, then P, Q, R and the quartic $\Delta = Q^2 - 4PR$ of Theorem Y.

Step 3: the elliptic coordinate on the circle. By Theorem Y, along the curve $F(\varphi, \psi) = 0$ we have $d\psi/\sqrt{\Delta(\psi)} = d\varphi/\sqrt{\Delta(\varphi)}$: in the coordinate $t = \int d\varphi/\sqrt{\Delta(\varphi)}$ the passage to the next vertex is a translation (Lemma 5). For $\varphi = e^{i\theta} \in \Gamma$, the points of the curve with real vertices form a closed loop of the torus, along which t moves in a *fixed* direction (numerically, the argument of $ie^{i\theta}/\sqrt{\Delta(e^{i\theta})}$ is constant modulo π). So we may use the real parameter

$$\tau(\theta) = \int_0^\theta \frac{d\theta'}{\left|\Delta(e^{i\theta'})\right|^{1/2}}, \qquad T = \tau(2\pi) \ \text{(real period)}, \qquad \omega = \tau(\theta_1) - \tau(\theta_0) \ \text{(Poncelet step)},$$

where $e^{i\theta_1}$ is the counterclockwise neighbour of $e^{i\theta_0}$ (a root of $F(e^{i\theta_0}, \psi) = 0$). The step ω does not depend on θ_0: this is Poncelet's theorem in the elliptic coordinate. The integrand is smooth and periodic, so these integrals are computed to machine precision.

Step 4: the closing condition. By Lemma 5(3), the polygons close after n steps, going once around Γ, if and only if

$$\frac{\omega}{T} = \frac{1}{n}.$$

The ratio ω/T is a rotation number. When $\sigma \to 0$, γ shrinks to a point, the sides become chords through that point and $\omega/T \to \frac{1}{2}$; when σ increases until γ touches Γ, the steps near the point of contact become arbitrarily small and $\omega/T \to 0$. By continuity, *for every shape and every* $n \geq 3$ *there is a scale* σ *with* $\omega/T = 1/n$. Here a bisection gives $\sigma = 0.9076589029$, with $T = 0.23868159$ and $\omega = 0.04773632$.

Step 5: the vertices from the elliptic function. The inverse function $f(\tau) = e^{i\theta(\tau)}$ of $\tau(\theta)$ is the elliptic function of Lemma 5, restricted to the real loop: it satisfies $(f')^2 = c\,\Delta(f)$ for a constant c. The vertices of the pentagons are $z_j = f(\tau_0 + j\omega)$, $j = 0, \ldots, 4$: in the coordinate τ they are equally spaced (Figure 2).

Step 6: the Blaschke product from Theorem X. By the elliptic form of Theorem X and (4), for each pentagon

$$N(z) - \lambda D(z) = \prod_{j=0}^{4} (z - z_j), \qquad \lambda = (-1)^{n+1} \prod_{j=0}^{4} z_j,$$

where $B = N/D$ with $N(z) = z^5 + c_4 z^4 + \cdots + c_1 z$ and $D(z) = 1 + d_1 z + \cdots + d_4 z^4$. Comparing coefficients gives 4 linear equations in the 8 unknowns c_k, d_k for each pentagon, so two pentagons ($\tau_0 = 0$ and $\tau_0 = 0.37\,\omega$) determine B. No conic is fitted at any point of the construction.

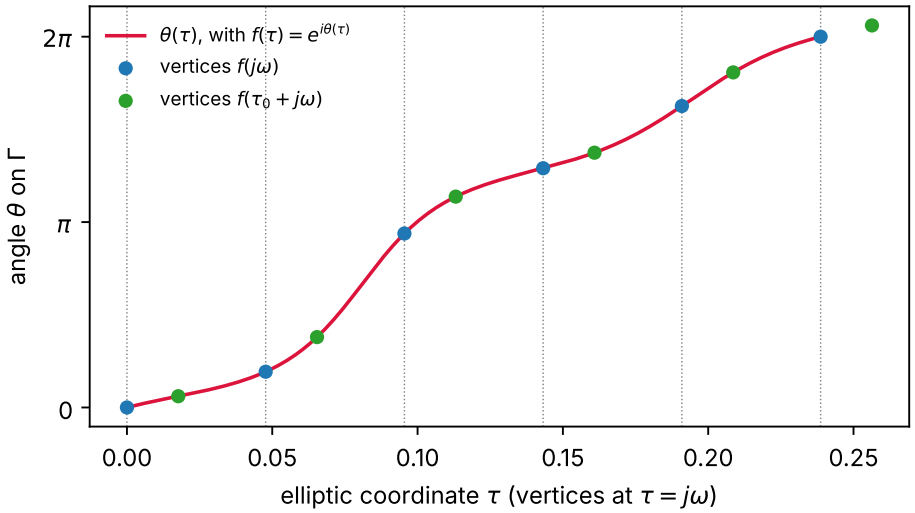


Figure 2: The elliptic function $f(\tau) = e^{i\theta(\tau)}$ for the degree-5 example. The vertices of a Poncelet pentagon are the values of f at five equally spaced points $\tau_0 + j\omega$; after 5 steps, θ has increased by 2π. Two pentagons are shown.

Checks. The script verifies, all to within 10^{-12} or better:

- ω computed from three different starting points is the same, and $f(5\omega) = f(0)$;

- the vertices $f(j\omega)$ coincide with those obtained geometrically by drawing tangents to γ;
- B is constant on five other pentagons, and the conic fitted afterwards to the sides of the polygons $B^{-1}(\lambda)$ is proportional to A^{-1}: we recover the ellipse we started from;
- the four critical points of B in $\mathbb{D}$ are simple, with two critical values: case (b) of Theorem Z;
- the zeros of B are 0, the foci of γ (roots of R, Lemma 2) and the foci of γ_2 (Theorem X);
- the four roots of Δ, i.e. the complex points of $\Gamma \cap \gamma$, are exactly the non-critical preimages of the critical values of B (remark on $\Gamma \cap \gamma$ in Section 5).

See Table 1 and Figure 3.

ellipse shape	centre $0.1200 - 0.1800\,i$, axis ratio 0.6, angle 30°
scale σ (so that $\omega/T = 1/5$)	0.9076589029
real period T, step ω	0.23868159, 0.04773632
roots of Δ ($\Gamma \cap \gamma$)	$1.2209 + 0.2356\,i$, $0.7896 + 0.1524\,i$
	$-0.8696 - 1.0031\,i$, $-0.4934 - 0.5692\,i$
zeros of B	$-0.5088 - 0.5431\,i$
	$-0.1599 - 0.4510\,i$
	0
	$0.6286 + 0.0042\,i$
	$0.7488 + 0.1831\,i$
critical points in $\mathbb{D}$	$-0.3963 - 0.4807\,i$
	$0.6850 + 0.1220\,i$
	$0.3591 - 0.0001\,i$
	$-0.0843 - 0.2429\,i$
critical values in $\mathbb{D}$	$-0.015018 + 0.037355\,i$
	$0.020658 + 0.012425\,i$
	(case (b) of Theorem Z)
γ: centre	$0.1200 - 0.1800\,i$
γ: semi-axes, angle	0.9077, 0.5446, 30.0°
γ: foci	$-0.5088 - 0.5431\,i$, $0.7488 + 0.1831\,i$
γ_2: foci	$-0.1599 - 0.4510\,i$, $0.6286 + 0.0042\,i$
residual of the dual-conic fit	$5.0 \cdot 10^{-16}$
Theorem X: max. distance zeros/foci	$3.8 \cdot 10^{-15}$

Table 1: The degree-5 example, constructed from the elliptic function.

6.2 A Blaschke product of degree 4

Construction. Let $D(z) = z\frac{a-z}{1-\bar{a}z}$ with $a = 0.5{+}0.3\,i$ and $C_1(w) = w\frac{s_1-w}{1-\bar{s}_1 w}$ with $s_1 = -0.4{+}0.25\,i$, and $B_4 = C_1 \circ D$. By Fujimura's theorem [4, 5], B_4 is associated to an ellipse.

Result. Table 2 and Figure 4.

- The critical points of B_4 are the critical point of D and the two preimages under D of the critical point of C_1. The latter two have the same value, so B_4 has two critical values and three simple critical points: case (b) of Theorem Z.
- The zeros of B_4 are 0, a (the zeros of D) and the two preimages $D^{-1}(s_1)$. Theorem X identifies them: $D^{-1}(s_1)$ are the foci of γ, and $a = P_2$ is the point through which all diagonals of the

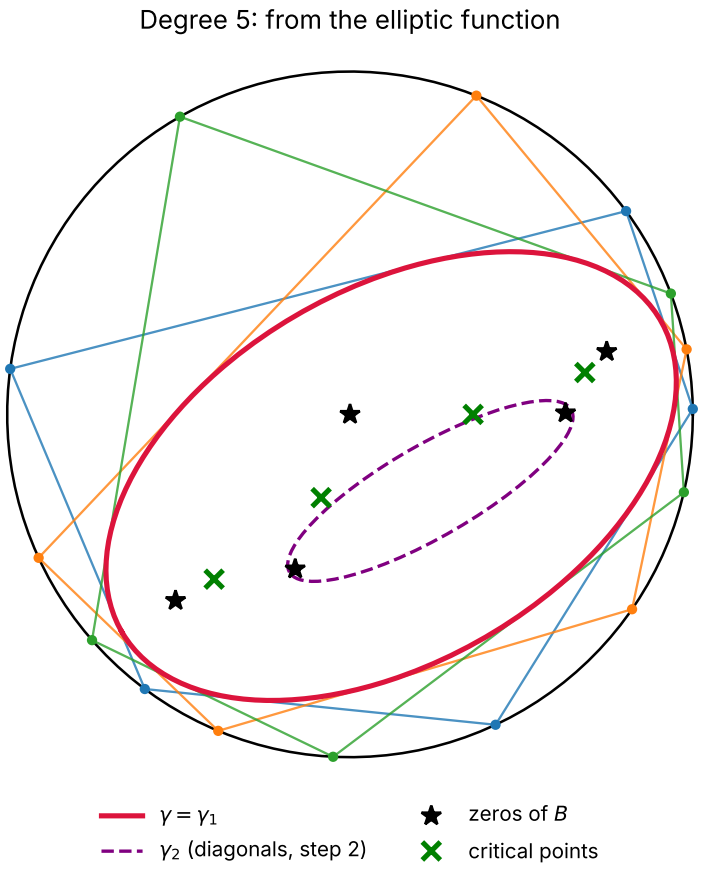


Figure 3: Degree 5: three Poncelet pentagons, the ellipse γ we started from, and the ellipse γ_2 enveloping the diagonals.

quadrilaterals pass. This is Lemma 4(2). Indeed, the quadrilateral $B_4^{-1}(\lambda)$ is the union of the two pairs $D^{-1}(\mu_1)$, $D^{-1}(\mu_2)$, where $C_1^{-1}(\lambda) = \{\mu_1, \mu_2\}$; since D wraps Γ twice around itself, the two pairs alternate on Γ, so opposite vertices z, z' satisfy $D(z) = D(z')$, i.e. $z' = \frac{a-z}{1-\bar{a}z}$ (the involution exchanging the two points of each fibre of D, which exchanges 0 and a), and the chord zz' passes through a.

construction	$D(z) = z\frac{a-z}{1-\bar{a}z}$, $a = 0.50 + 0.30\,i$
	$C_1(w) = w\frac{s_1-w}{1-\bar{s}_1 w}$, $s_1 = -0.40 + 0.25\,i$
zeros of B	$-0.4252 + 0.3931\,i$
	0
	$0.5000 + 0.3000\,i$
	$0.8002 + 0.1519\,i$
critical points in $\mathbb{D}$	$-0.2470 + 0.2578\,i$
	$0.6806 + 0.1724\,i$
	$0.2759 + 0.1655\,i$
critical values in $\mathbb{D}$	$0.027534 - 0.056481\,i$
	$-0.038009 - 0.031506\,i$
	(case (b) of Theorem Z)
γ: centre	$0.1875 + 0.2725\,i$
γ: semi-axes, angle	0.7578, 0.4292, $-11.1°$
γ: foci	$-0.4252 + 0.3931\,i$, $0.8002 + 0.1519\,i$
P_2 (long diagonals)	$0.5000 + 0.3000\,i$
residual of the dual-conic fit	$4.3 \cdot 10^{-16}$
Theorem X: max. distance zeros/foci	$1.0 \cdot 10^{-15}$

Table 2: The degree-4 example $B_4 = C_1 \circ D$.

6.3 A Blaschke product of degree 8, from the Corollary

Construction. Let v_1, v_2 be the two critical values of B_4 (Table 2). Corollary 1 gives the unique $s \in \mathbb{D}$ for which $B_8 = C_2 \circ B_4$, with $C_2(w) = w\frac{s-w}{1-\bar{s}w}$, is associated to an ellipse:

$$s = \frac{v_1 + v_2 - v_1 v_2(\bar{v}_1 + \bar{v}_2)}{1 - |v_1 v_2|^2},$$

which gives $s = -0.0104 - 0.0877\,i$. As a control, replacing s by $s + 0.1$ gives a residual of $6.8 \cdot 10^{-3}$: the sides of the octagons are then not tangent to a common conic.

Result. Table 3 and Figure 5.

- B_8 has seven simple critical points in $\mathbb{D}$ and two critical values: case (b) of Theorem Z. Three of the critical points are those of B_4; the four others are the preimages under B_4 of the critical point of C_2. The choice of s makes $C_2(v_1) = C_2(v_2)$, which merges the critical values coming from B_4.
- The Poncelet package consists of $\gamma = \gamma_1$, γ_2, γ_3 and the point P_4, and the zeros of B_8 are 0, the six foci and P_4 (Theorem X).
- The ellipse γ_2 of B_8 is the ellipse γ of B_4 (same foci), and $P_4 = a$. Indeed, the octagon $B_8^{-1}(\lambda)$ is the union of the two quadrilaterals $B_4^{-1}(\mu_1)$, $B_4^{-1}(\mu_2)$, where $C_2^{-1}(\lambda) = \{\mu_1, \mu_2\}$; their vertices alternate (as B_4 wraps Γ four times around itself, passing alternately through

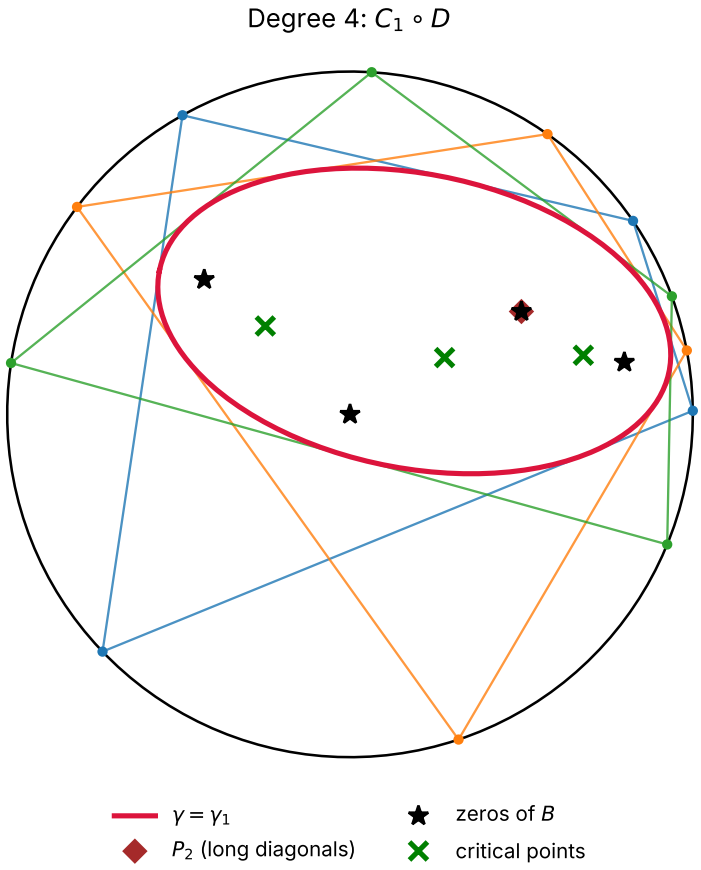


Figure 4: Degree 4: three Poncelet quadrilaterals, the ellipse γ, and the point $P_2 = a$ where the diagonals meet.

μ_1 and μ_2), so taking every second vertex of the octagon gives back the quadrilaterals of B_4. In the language of Section 5, the isogeny of degree 8 factors through the isogeny of degree 4 (Remark on decompositions).

construction	$C_2 \circ B_4$, $C_2(w) = w\frac{s-w}{1-\bar{s}w}$
	$s = \frac{v_1+v_2-v_1v_2(\bar{v}_1+\bar{v}_2)}{1-\lvert v_1v_2\rvert^2} = -0.0104 - 0.0877\,i$
zeros of B	$-0.4548 + 0.2291\,i$
	$-0.4252 + 0.3931\,i$
	$-0.0037 + 0.3821\,i$
	0
	$0.5000 + 0.3000\,i$
	$0.5614 + 0.0290\,i$
	$0.7744 + 0.2290\,i$
	$0.8002 + 0.1519\,i$
critical points in $\mathbb{D}$	$-0.4133 + 0.3127\,i$
	$0.7744 + 0.1837\,i$
	$0.6806 + 0.1724\,i$
	$0.5136 + 0.1614\,i$
	$-0.2470 + 0.2578\,i$
	$0.2759 + 0.1655\,i$
	$0.0015 + 0.1993\,i$
critical values in $\mathbb{D}$	$-0.001904 + 0.000458\,i$
	$-0.002826 + 0.001279\,i$
	(case (b) of Theorem Z)
γ: centre	$0.0533 + 0.1290\,i$
γ: semi-axes, angle	$0.9357,\ 0.7793,\ -11.1°$
γ: foci	$-0.4548 + 0.2291\,i,\ 0.5614 + 0.0290\,i$
γ_2: foci	$-0.4252 + 0.3931\,i,\ 0.8002 + 0.1519\,i$
γ_3: foci	$-0.0037 + 0.3821\,i,\ 0.7744 + 0.2290\,i$
P_4 (long diagonals)	$0.5000 + 0.3000\,i$
residual of the dual-conic fit	$7.8 \cdot 10^{-16}$
Theorem X: max. distance zeros/foci	$2.8 \cdot 10^{-14}$

Table 3: The degree-8 example $B_8 = C_2 \circ B_4$, with s given by Corollary 1.

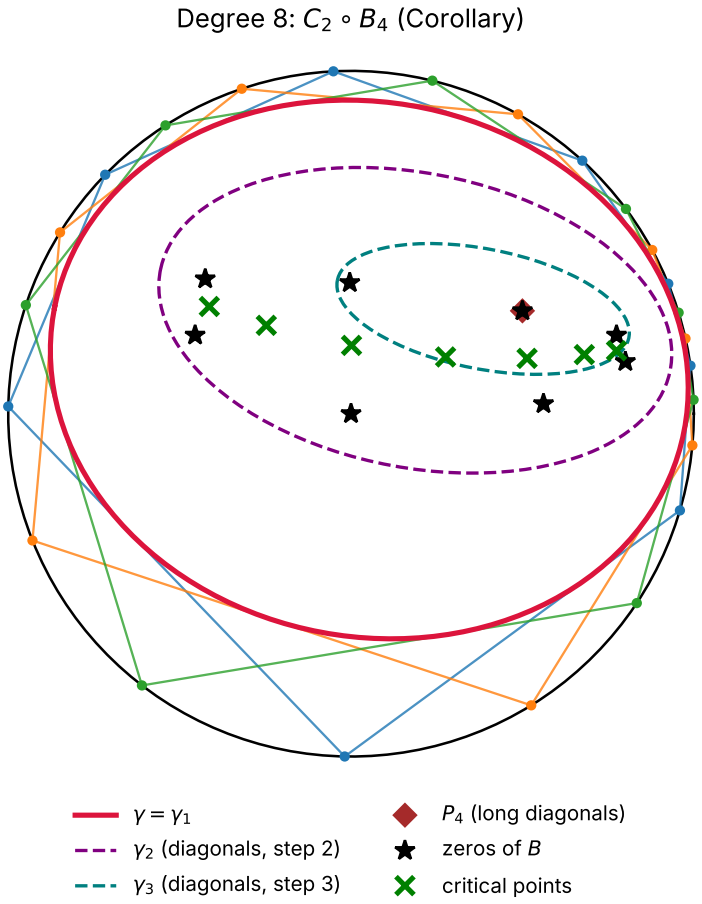


Figure 5: Degree 8: three Poncelet octagons, the ellipse γ, the ellipses γ_2 (which is the ellipse of B_4) and γ_3, and the point P_4.

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